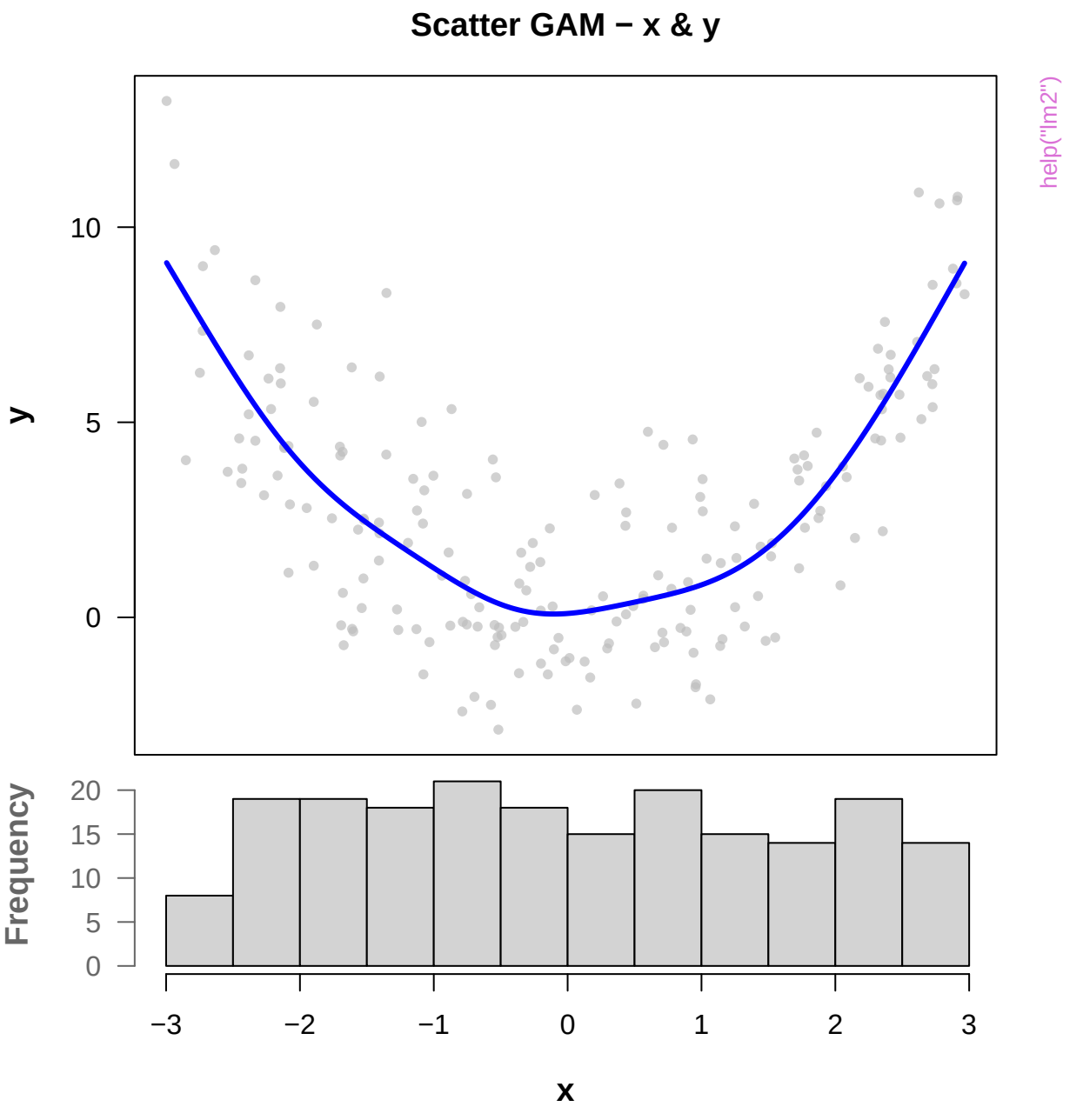
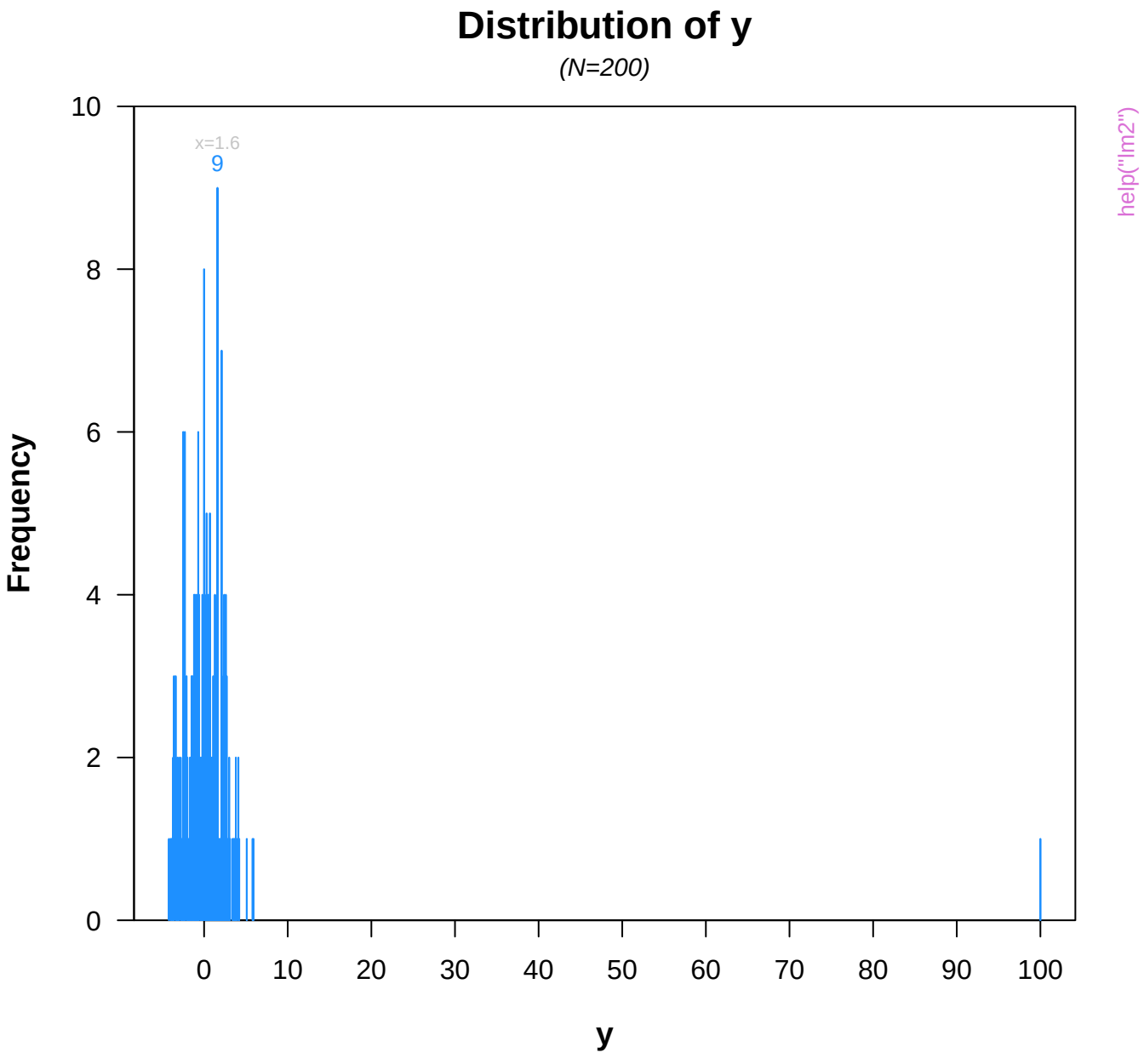
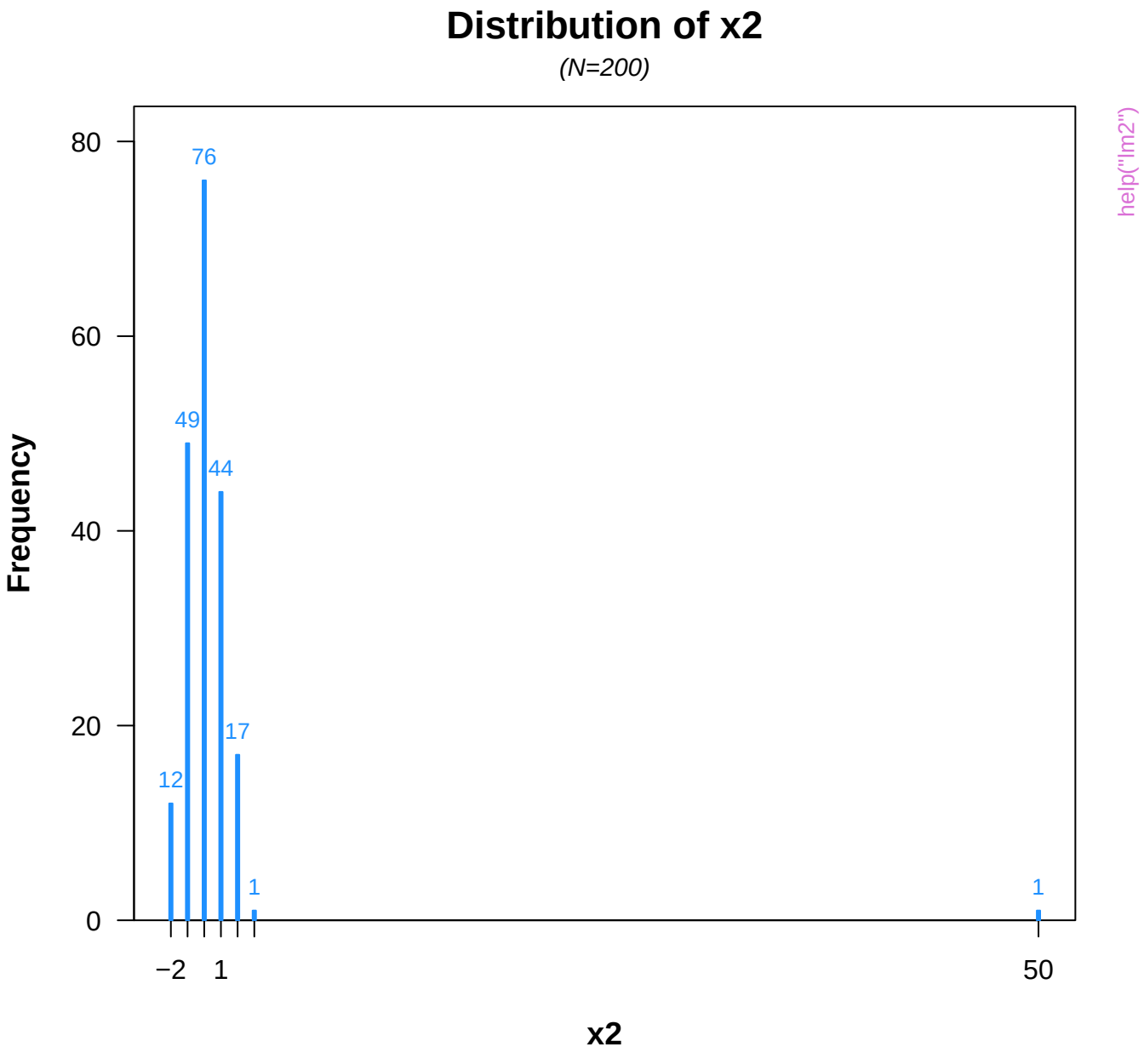
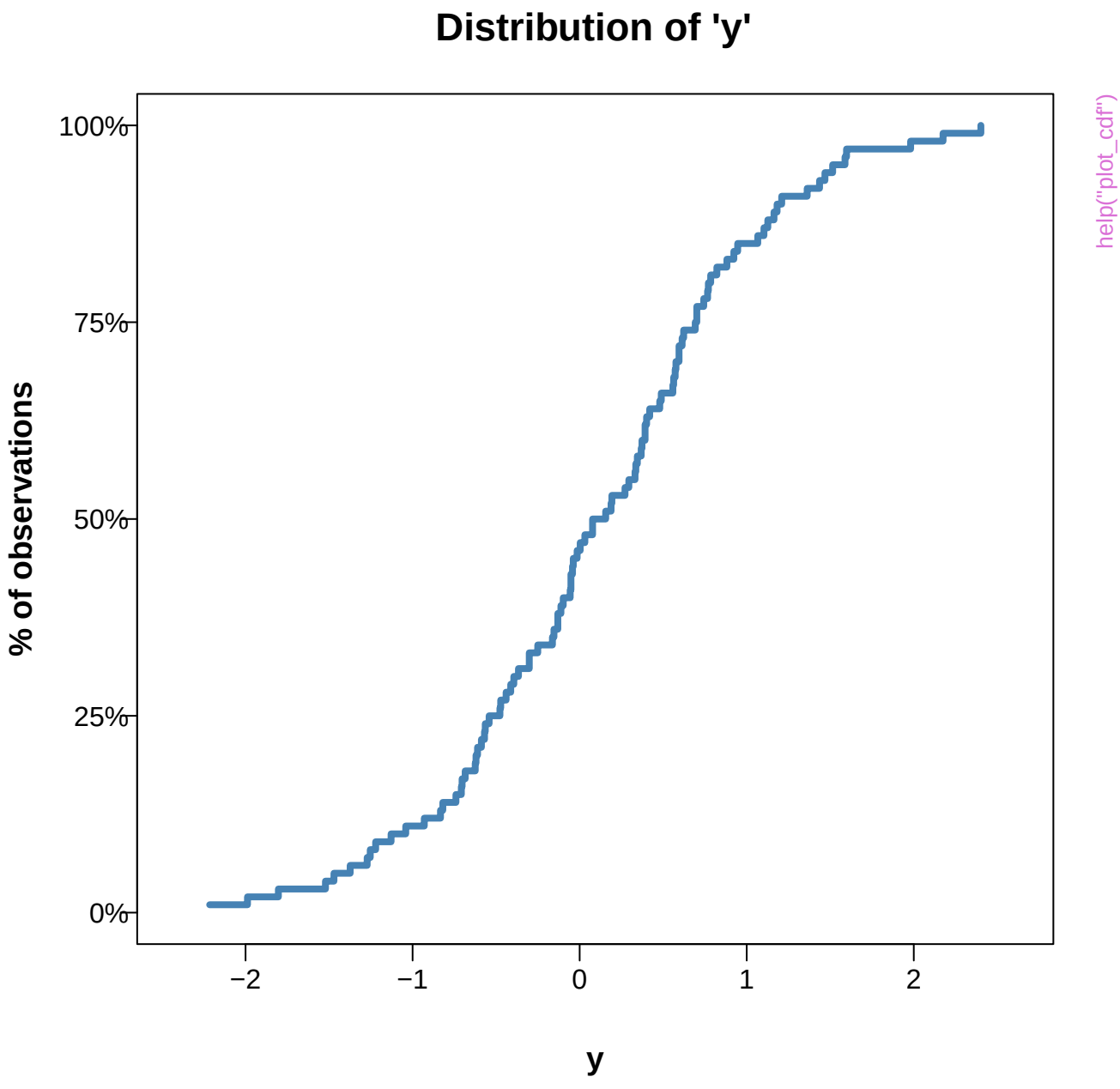
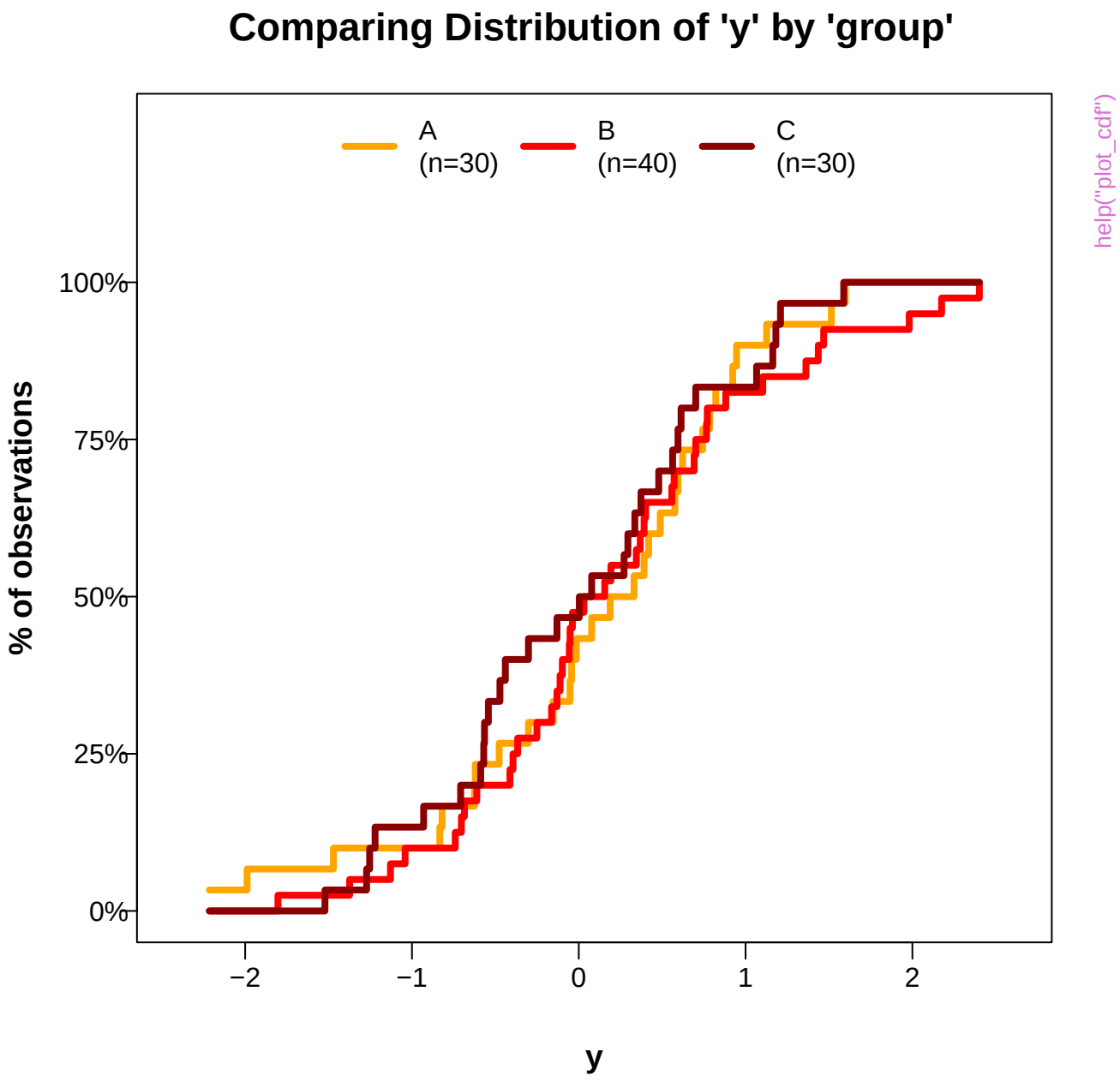
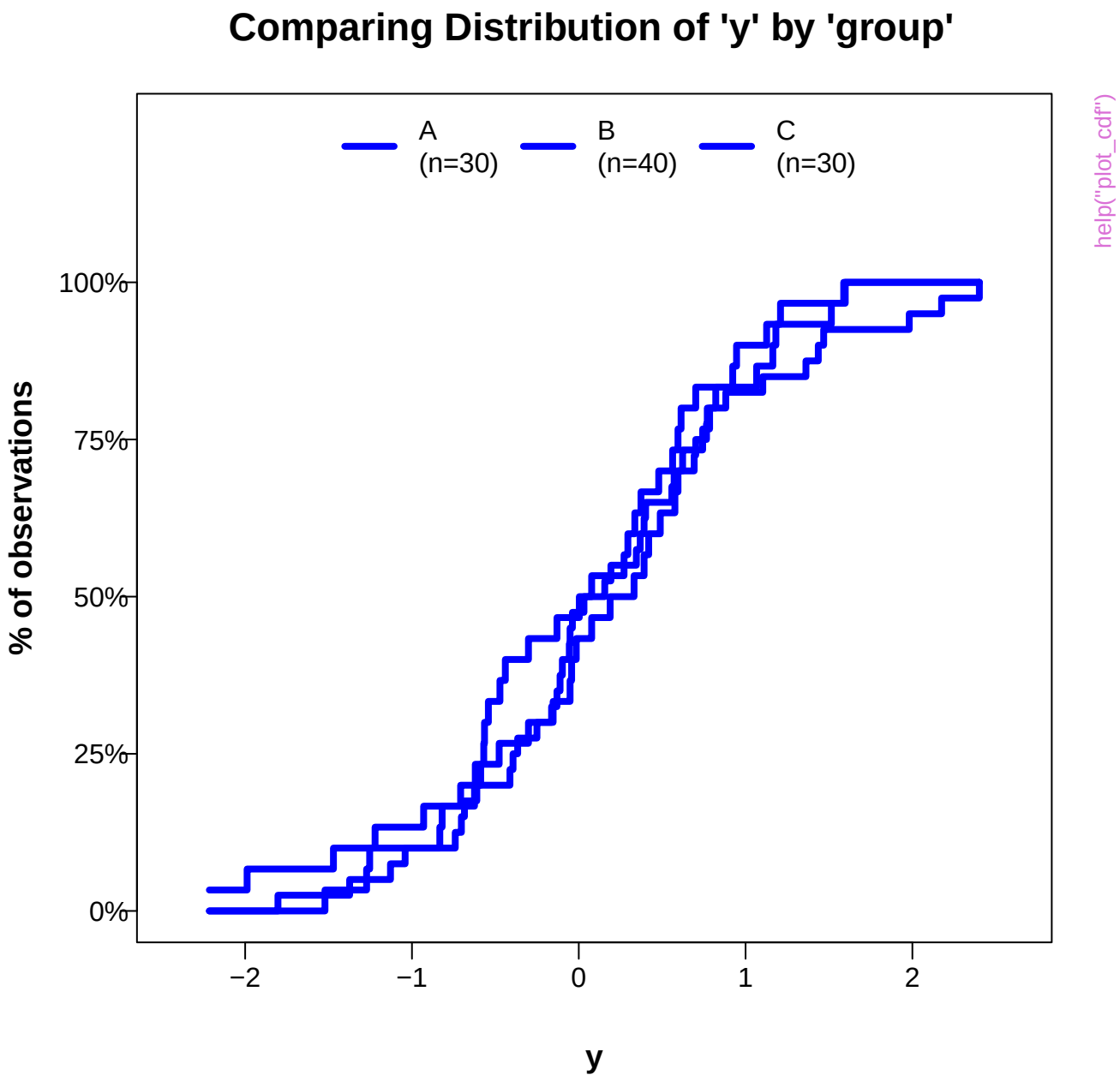

Scatter GAM − x & y
help("lm2")
y
10
5
0
Frequency
20
15
10
5
0
−3
−2
−1
0
1
2
3
x

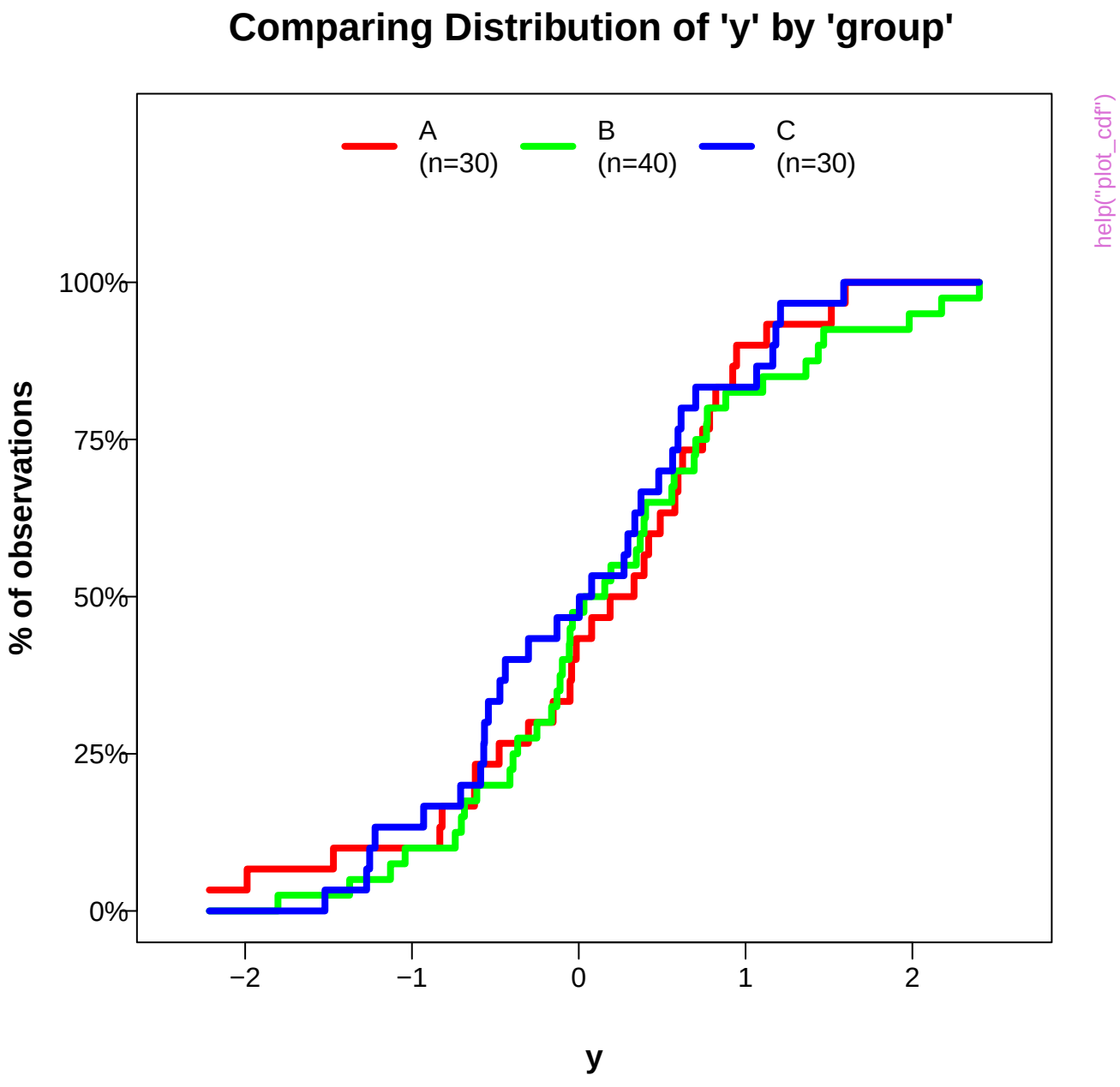

Distribution of y
(N=200)
x=1.6
9
Frequency
y
0
2
4
6
8
10
0
10
20
30
40
50
60
70
80
90
100
help("lm2")

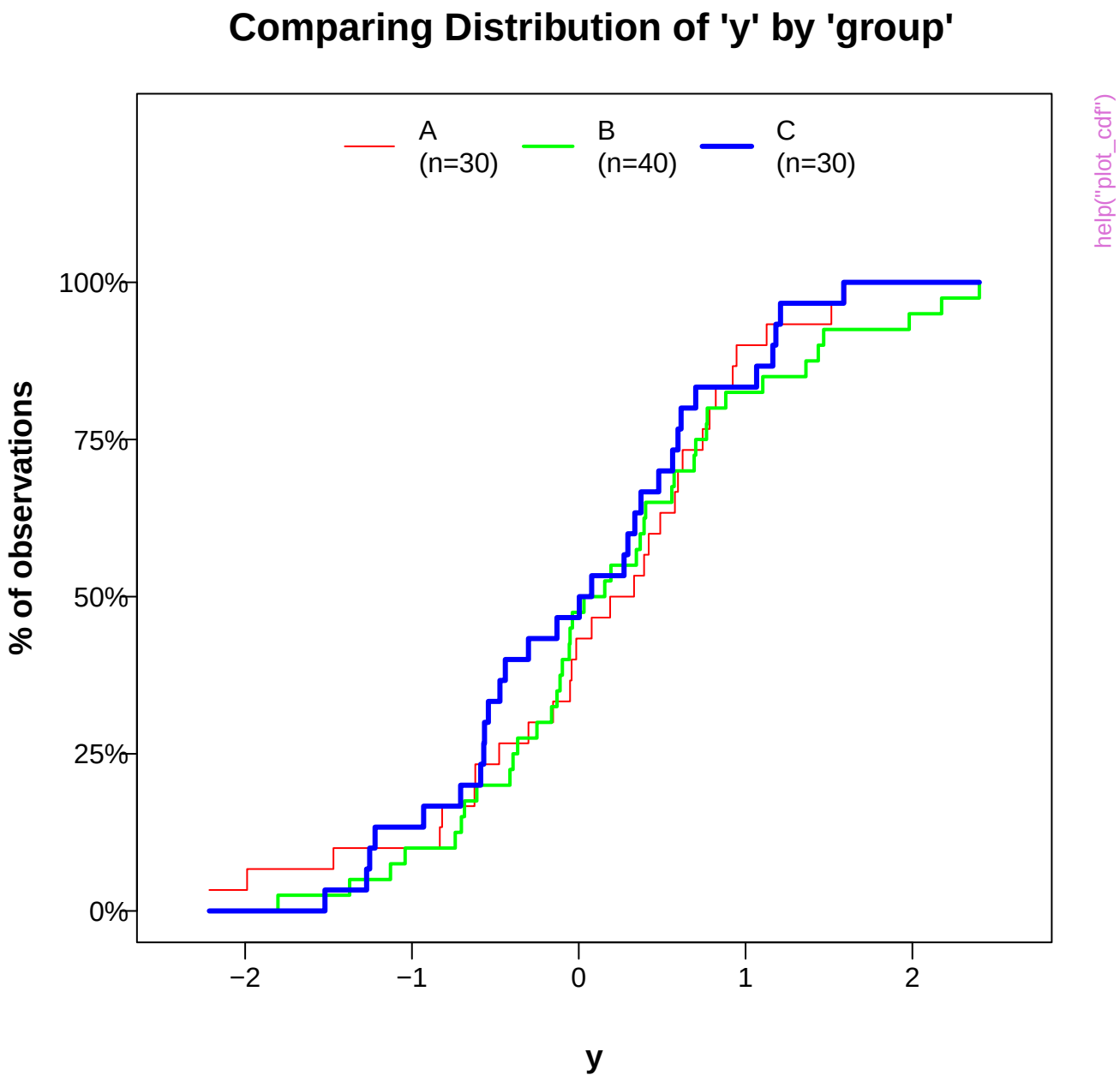

Distribution of x2
(N=200)
Frequency
80
60
40
20
0
76
49
44
17
12
1
1
−2
1
50
x2
help("lm2")

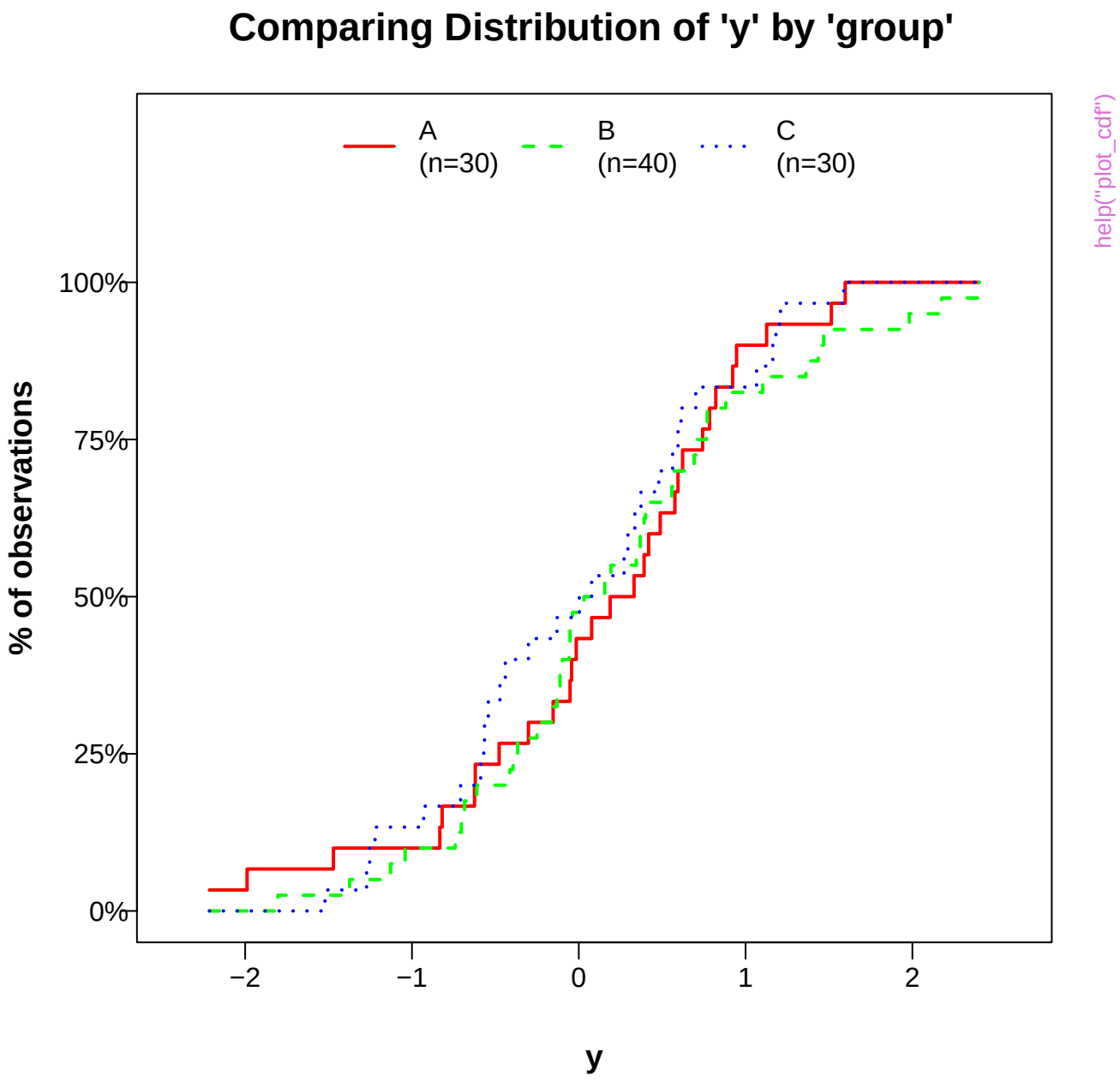

Distribution of 'y'
% of observations
100%
75%
50%
25%
0%
−2
−1
0
1
2
y
help("plot_cdf")

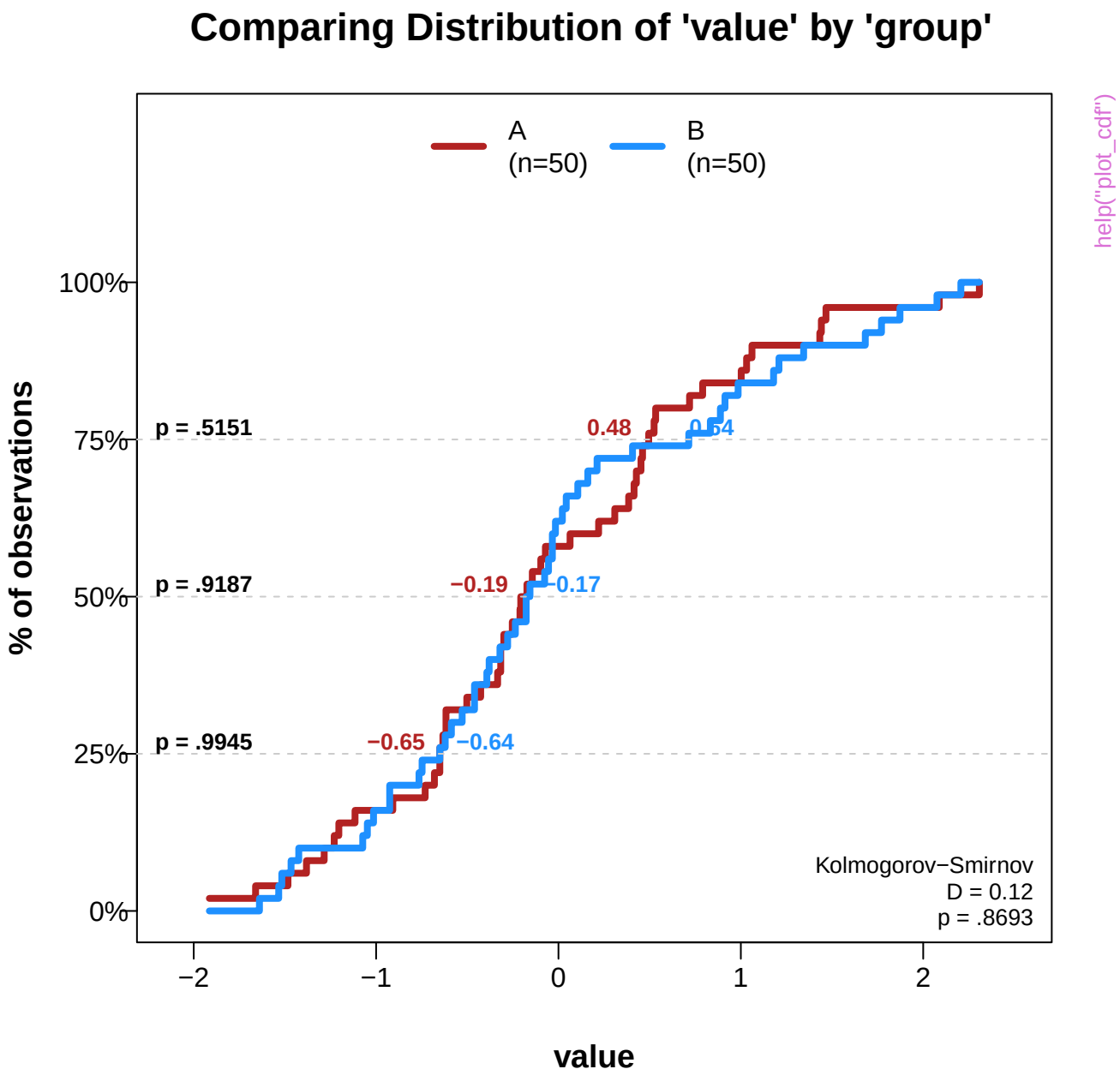

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

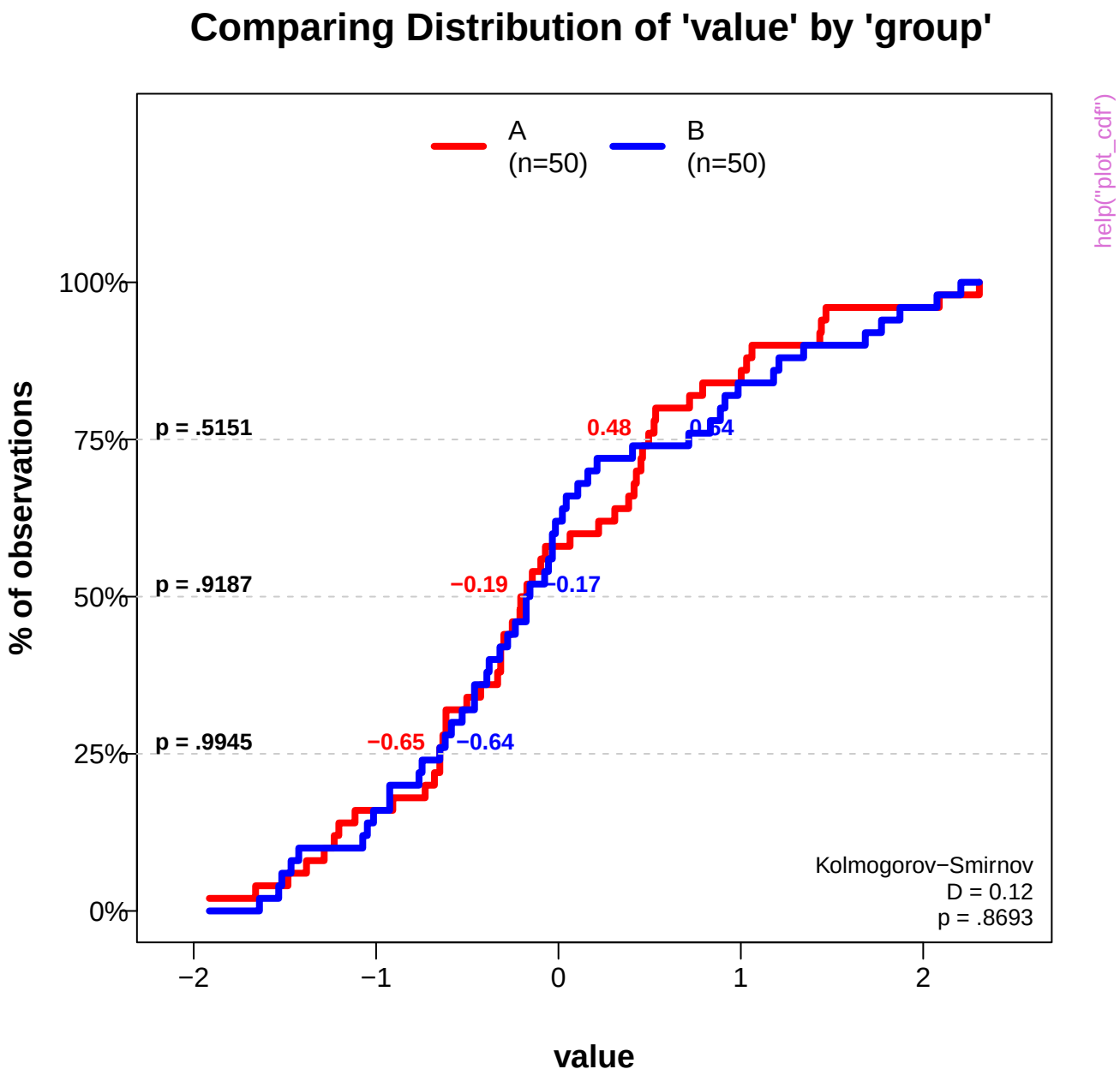

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

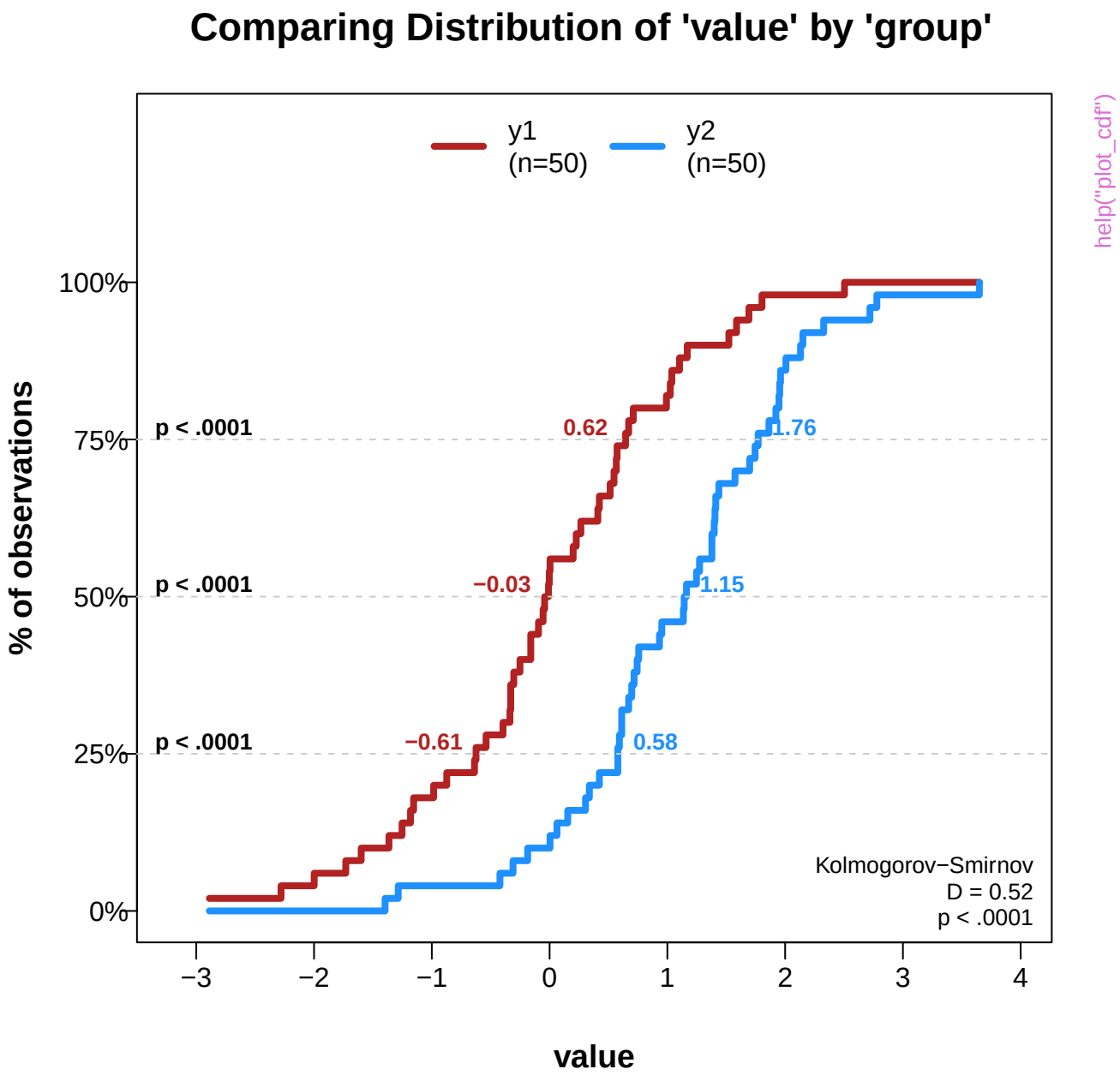

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
help("plot_cdf")
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y


Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")

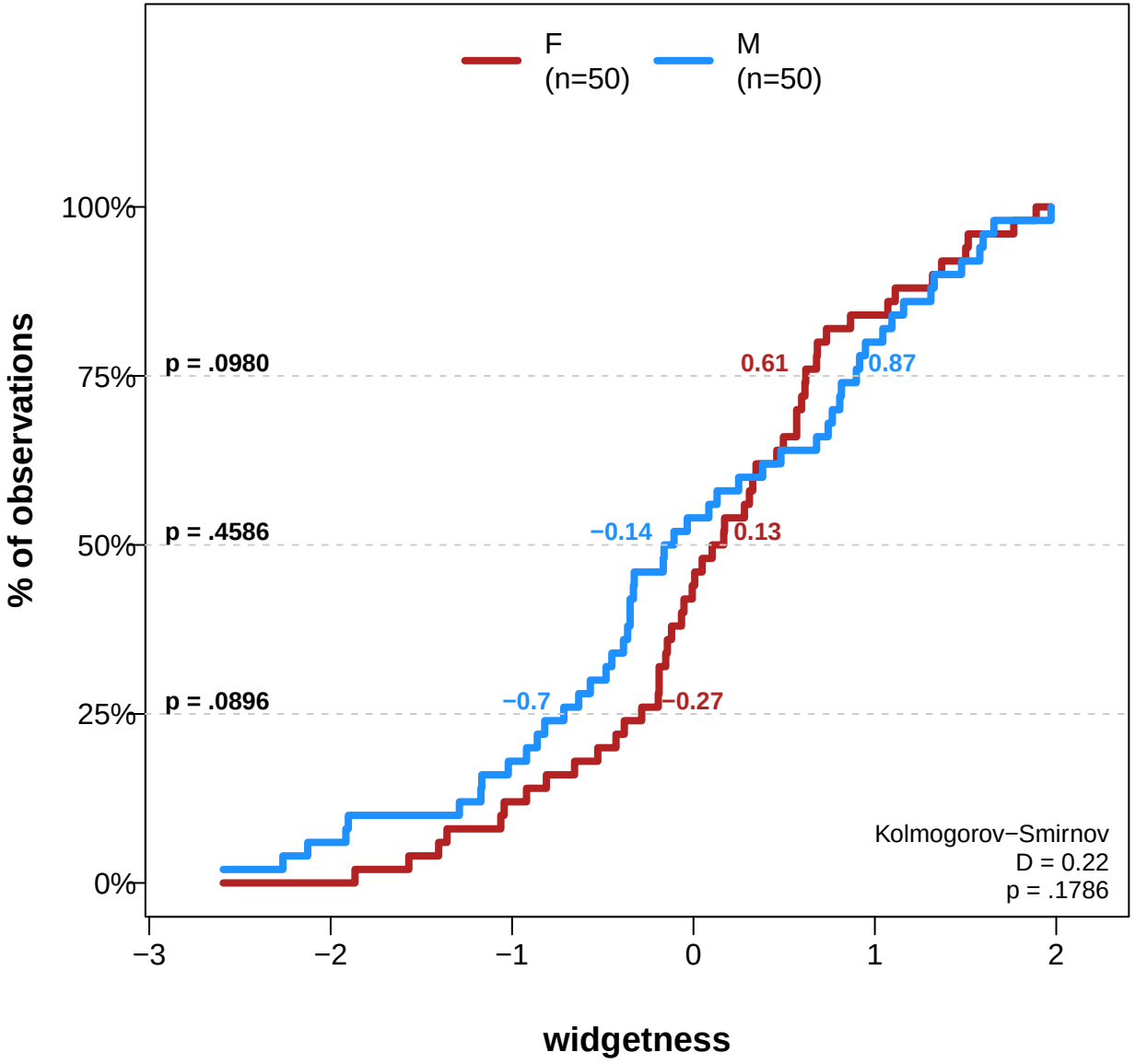

Comparing Distribution of 'y' by 'group'
A
(n=30)
B
(n=40)
C
(n=30)
100%
75%
50%
25%
0%
% of observations
−2
−1
0
1
2
y
help("plot_cdf")


Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5151
p = .9187
p = .9945
0.48
0.54
−0.19
−0.17
−0.65
−0.64
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value
help("plot_cdf")

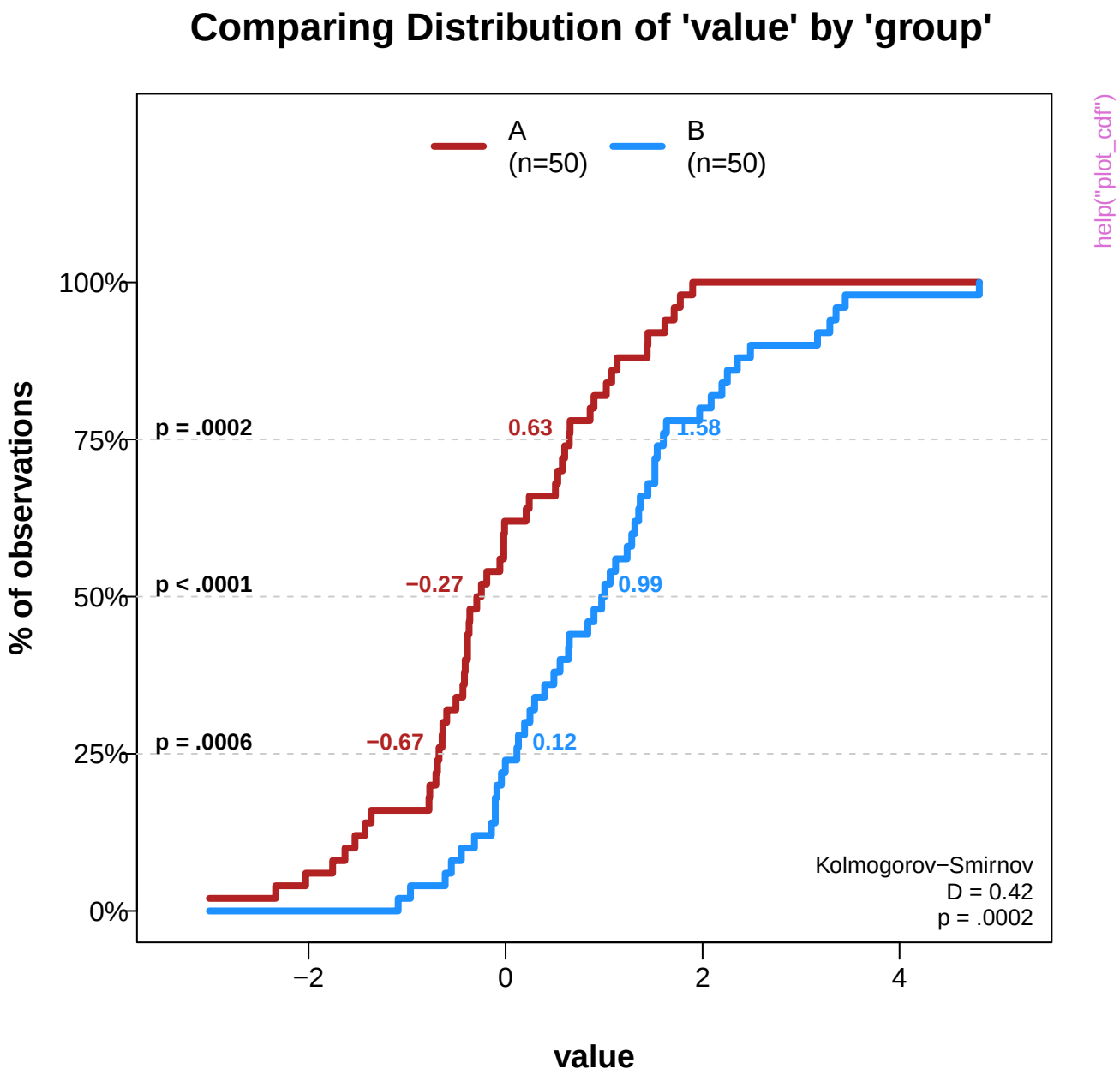

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .5151
p = .9187
p = .9945
0.48
0.54
−0.19
−0.17
−0.65
−0.64
Kolmogorov−Smirnov
D = 0.12
p = .8693
−2
−1
0
1
2
value
help("plot_cdf")

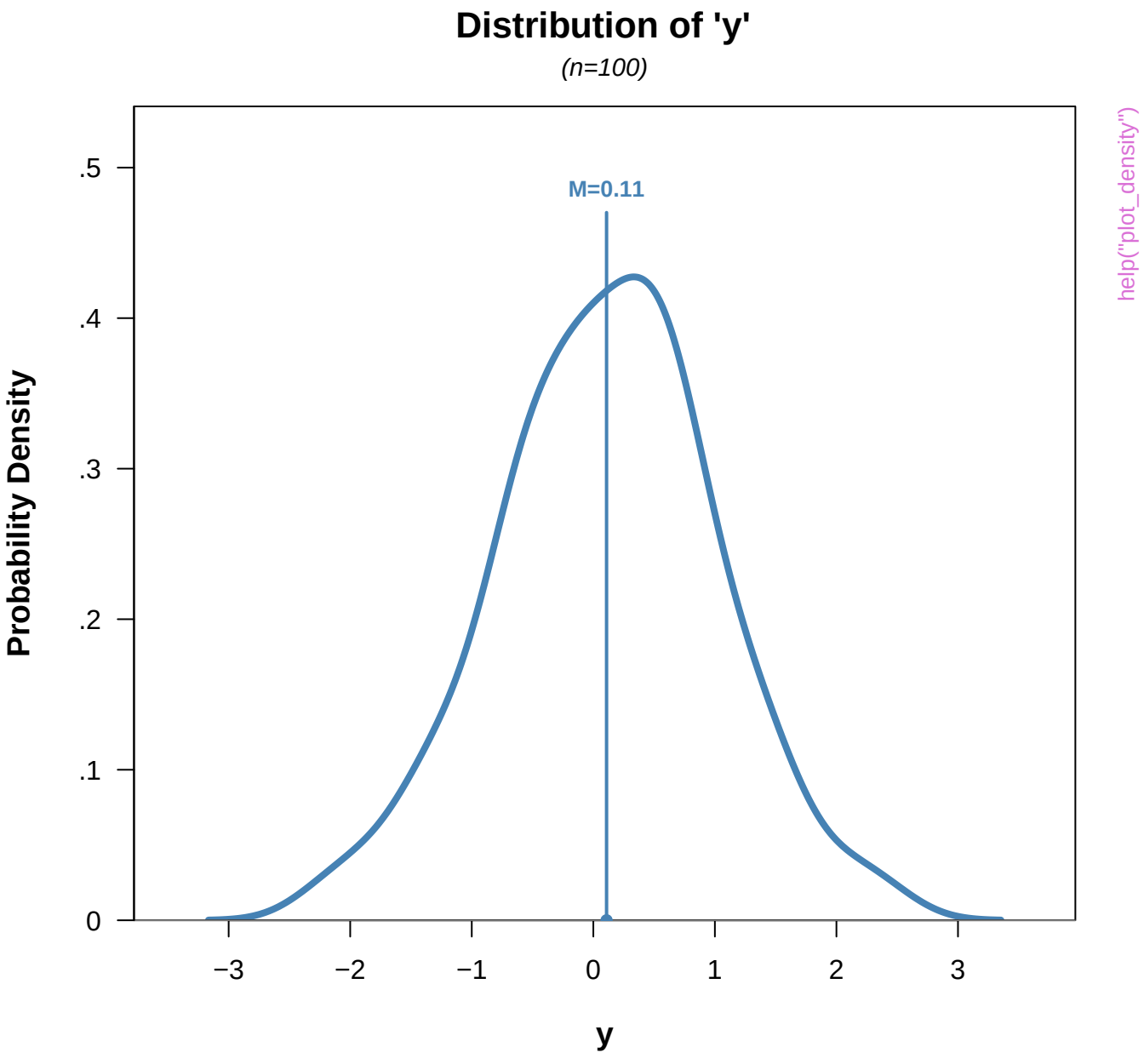

Comparing Distribution of 'value' by 'group'
y1
(n=50)
y2
(n=50)
100%
75%
50%
25%
0%
% of observations
p < .0001
p < .0001
p < .0001
0.62
−0.03
−0.61
1.76
1.15
0.58
Kolmogorov−Smirnov
D = 0.52
p < .0001
−3
−2
−1
0
1
2
3
4
value
help("plot_cdf")

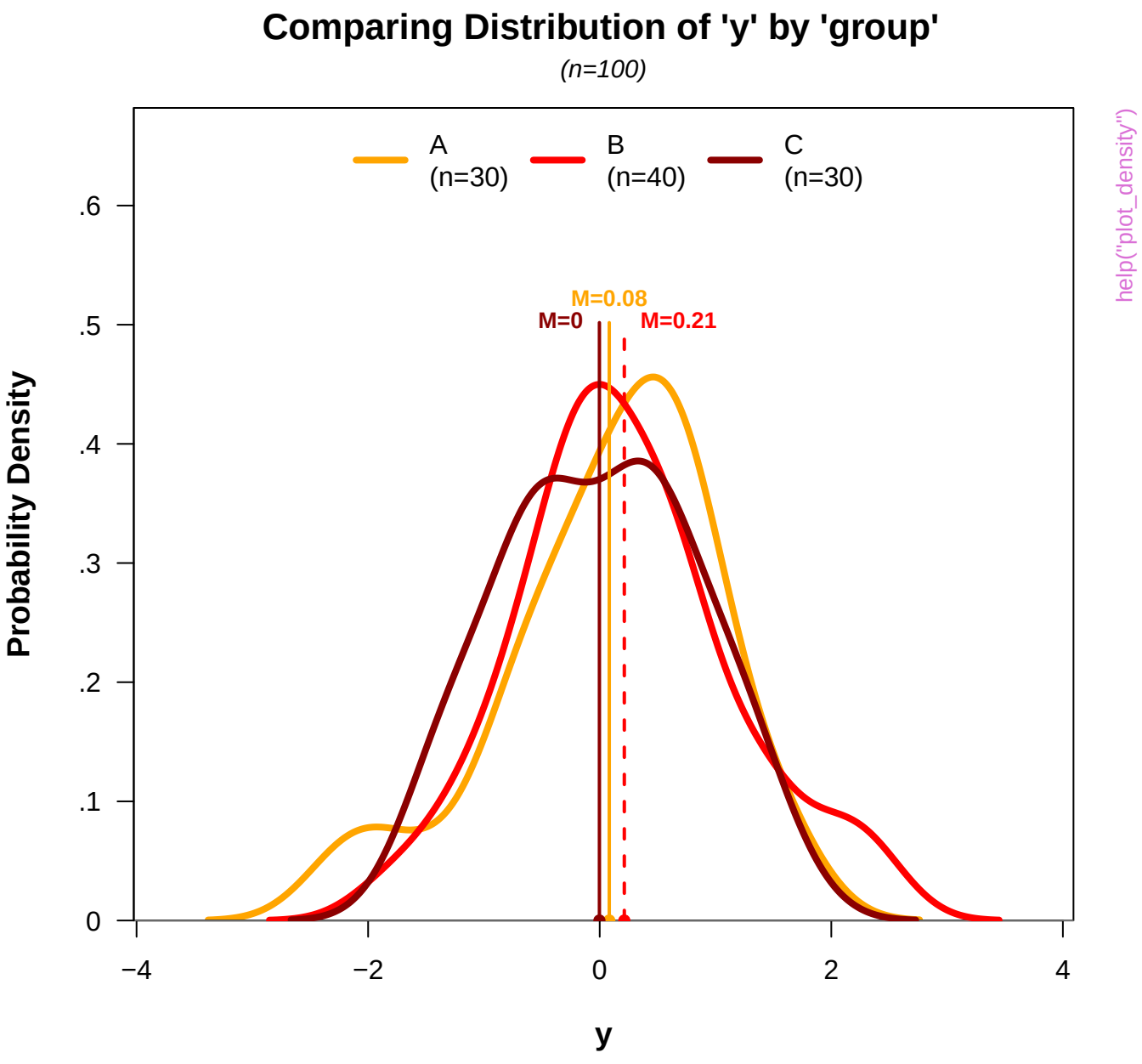
Comparing Distribution of 'widgetness' by 'gender'

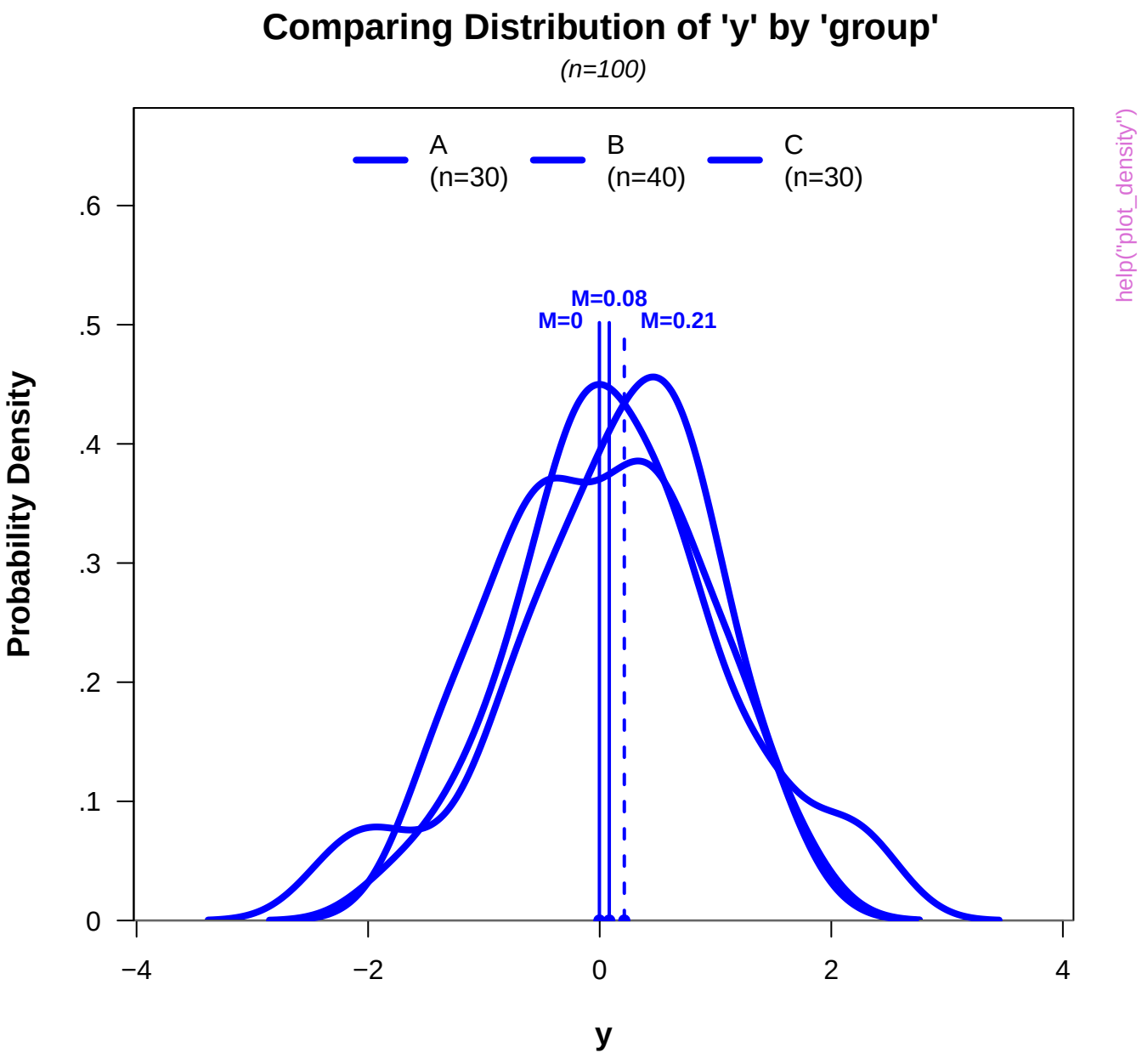


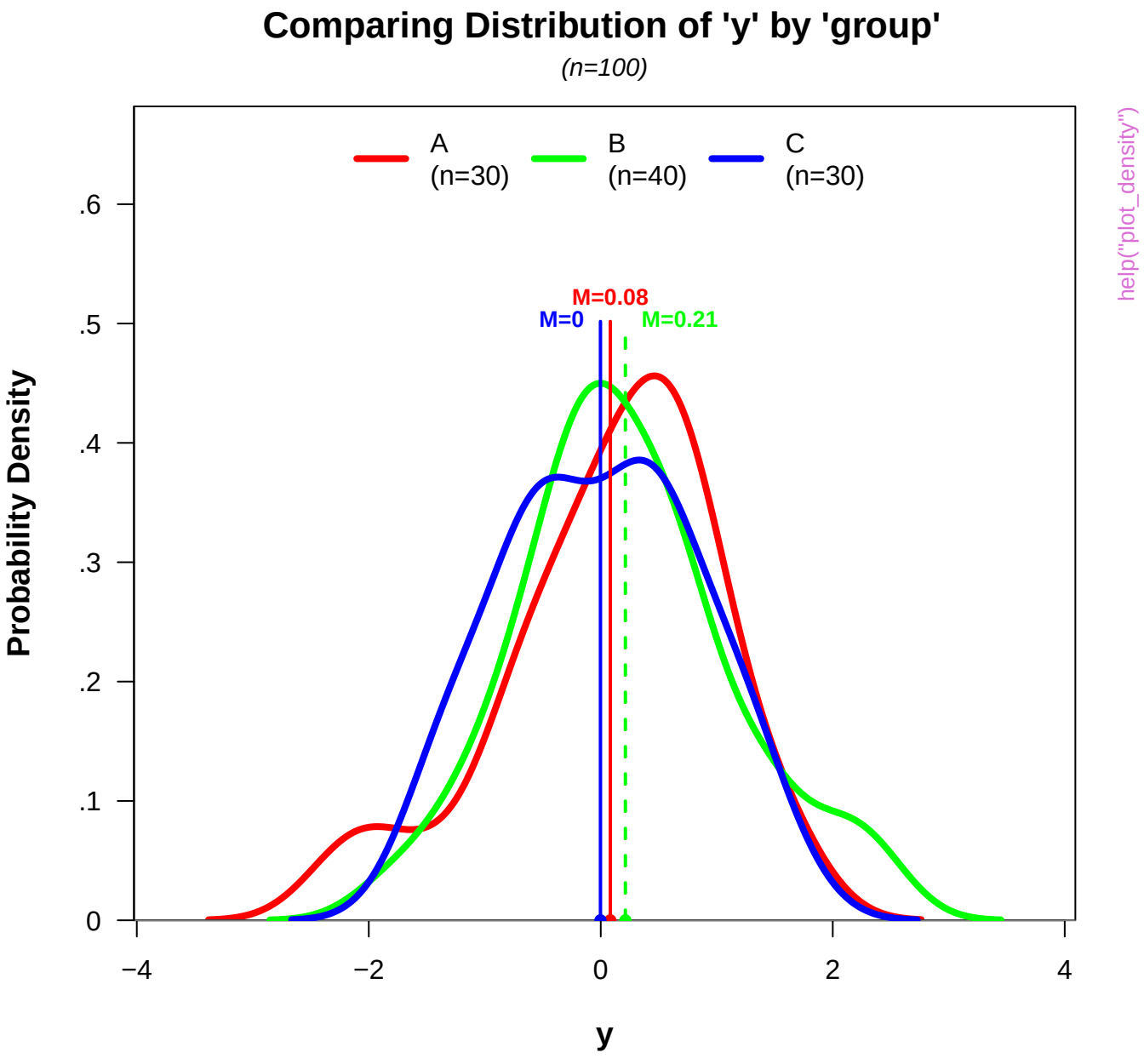
help("plot_cdf")

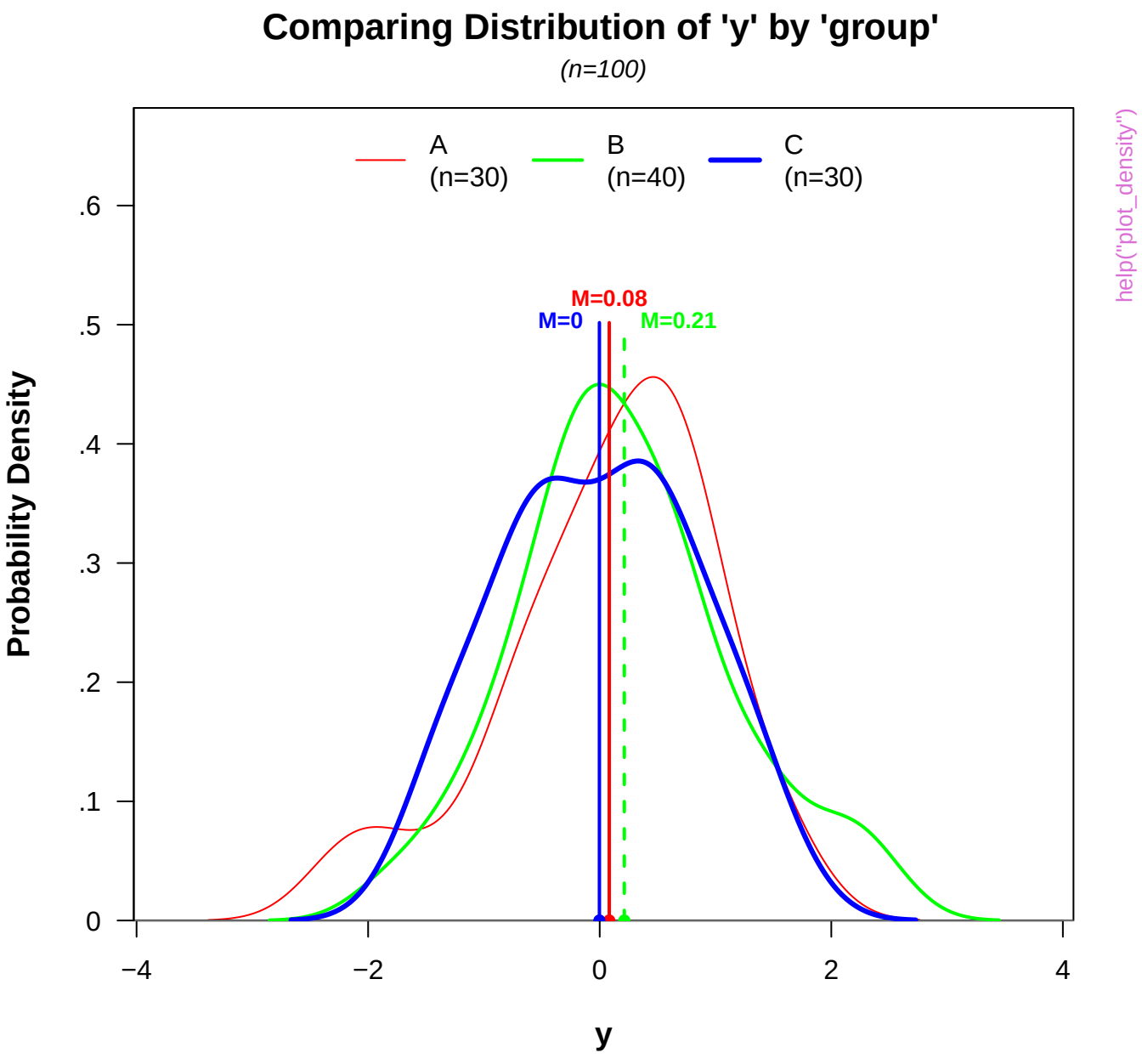

Comparing Distribution of 'value' by 'group'
A
(n=50)
B
(n=50)
100%
75%
50%
25%
0%
% of observations
p = .0002
p < .0001
p = .0006
0.63
−0.27
−0.67
1.58
0.99
0.12
Kolmogorov−Smirnov
D = 0.42
p = .0002
−2
0
2
4
value
help("plot_cdf")

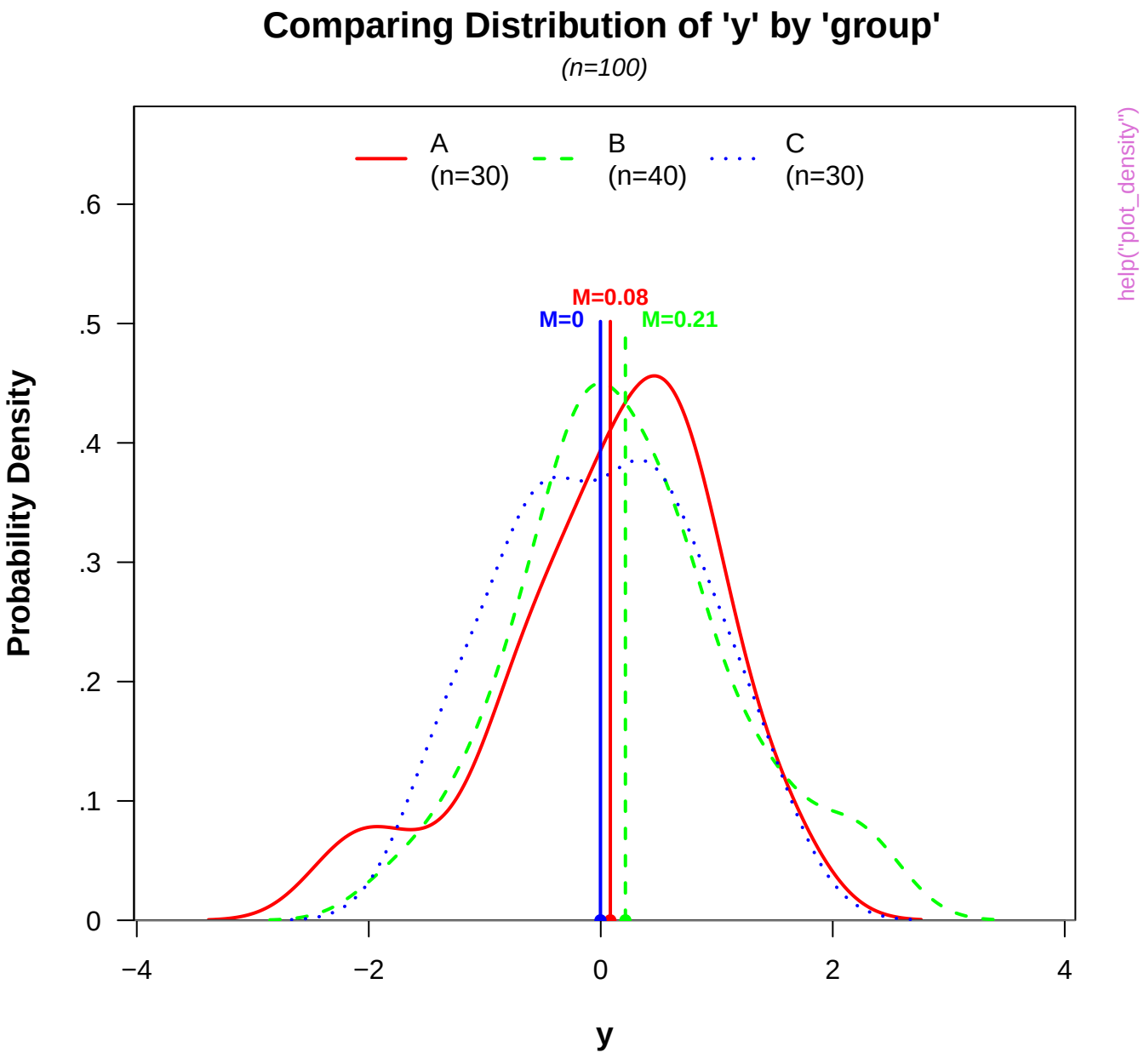

Distribution of 'y'
(n=100)
M=0.11
Probability Density
.5
.4
.3
.2
.1
0
−3
−2
−1
0
1
2
3
y
help("plot_density")

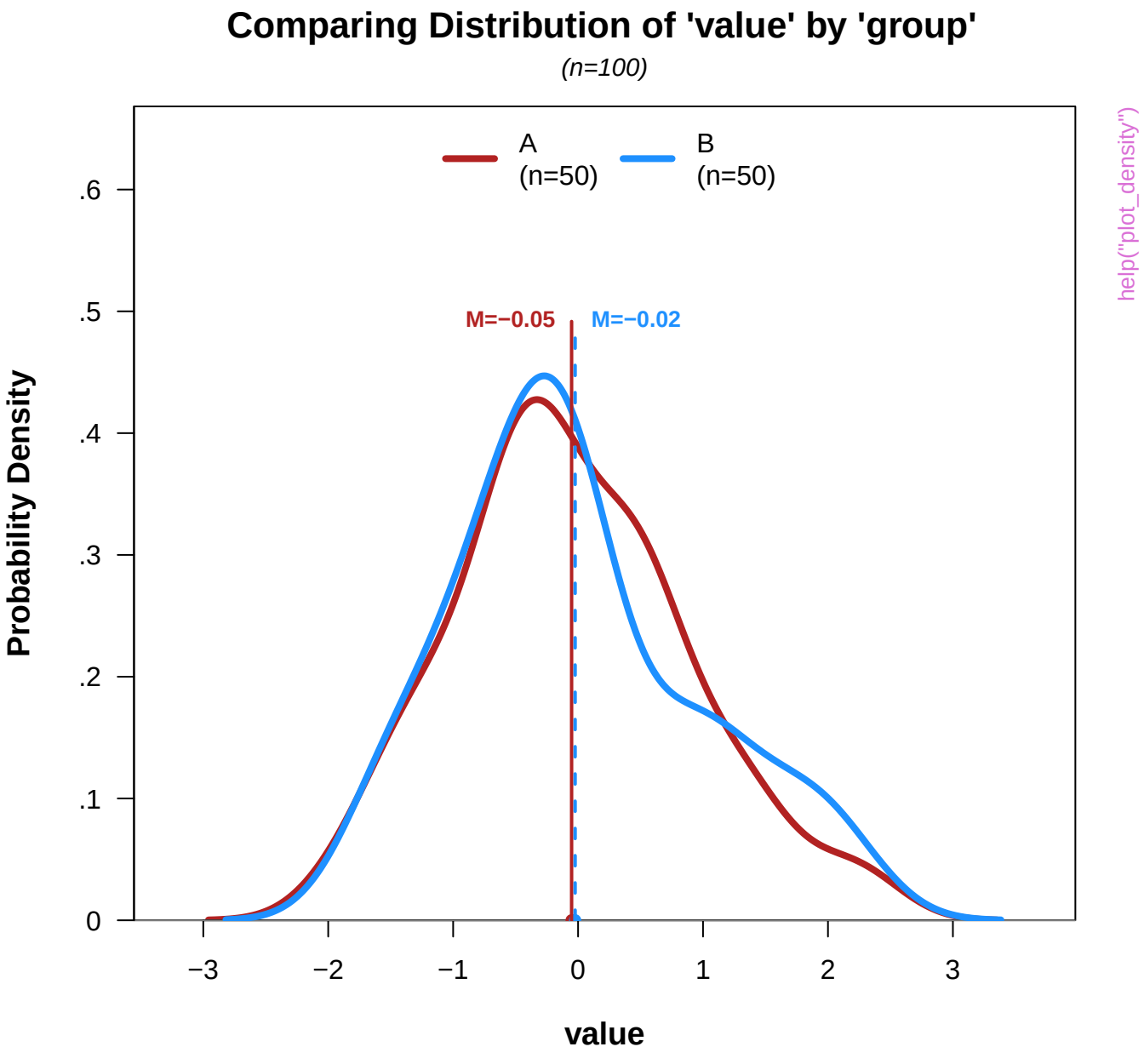

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

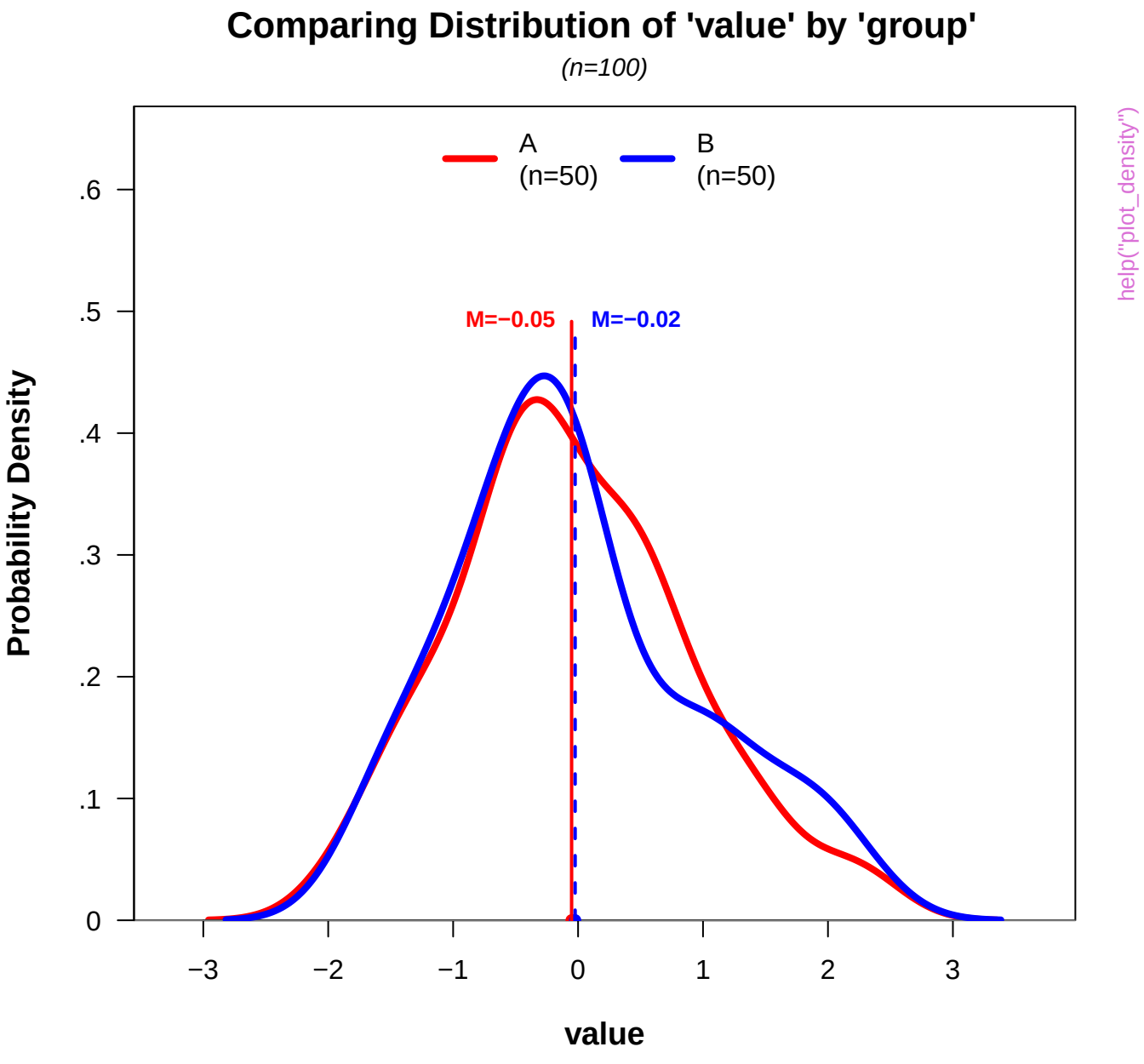

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

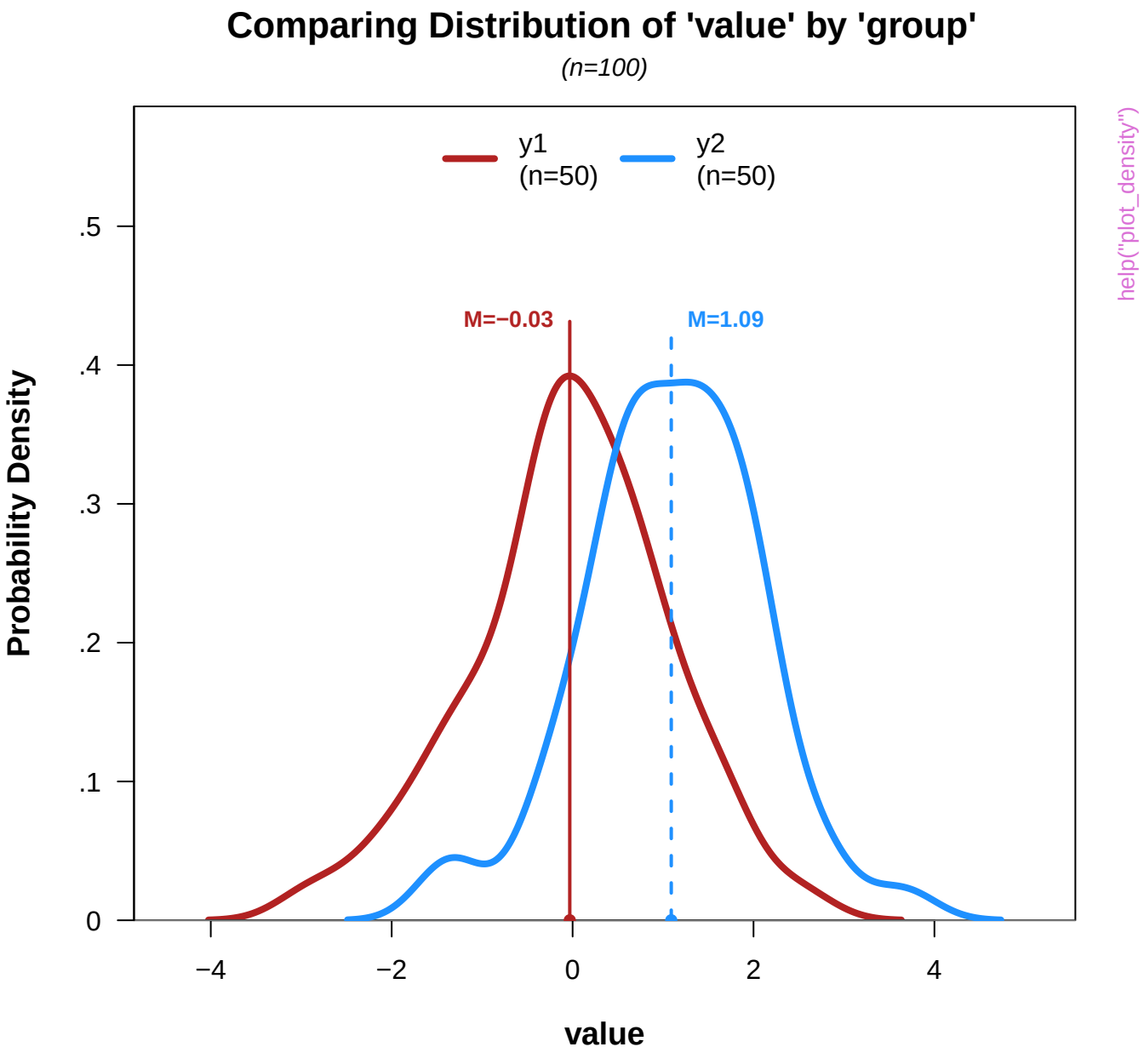

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

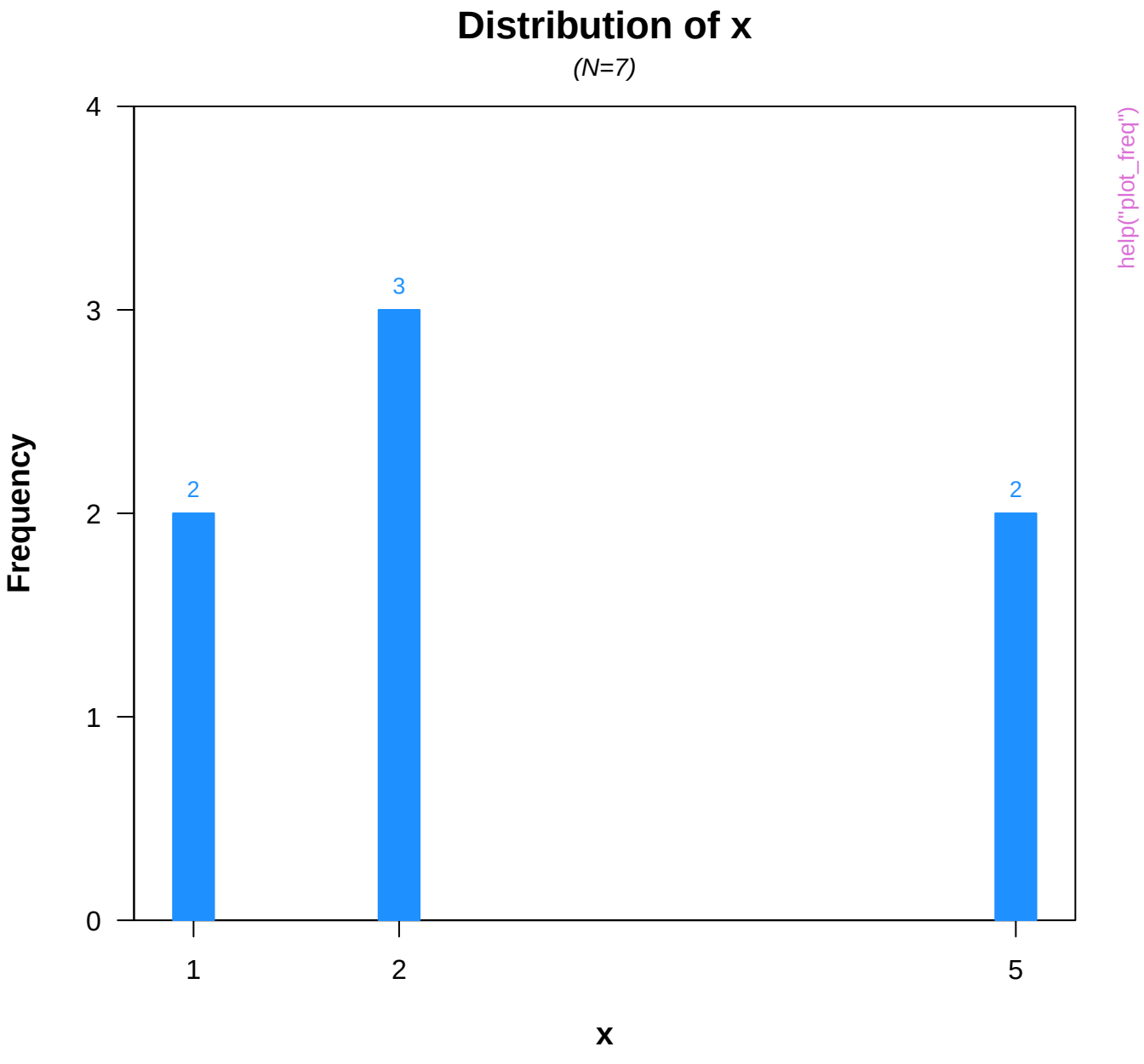

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

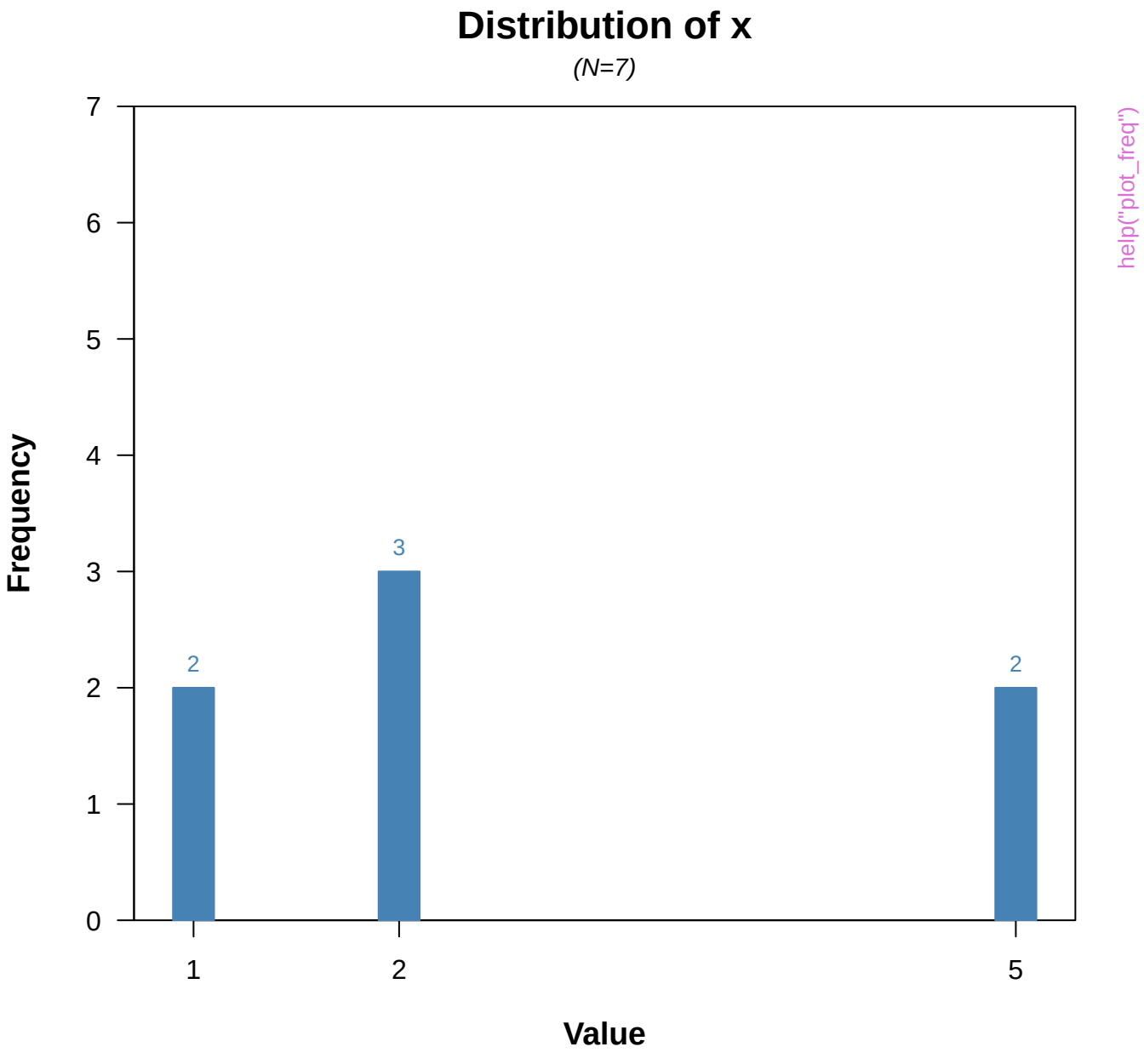

Comparing Distribution of 'y' by 'group'
(n=100)
A
(n=30)
B
(n=40)
C
(n=30)
M=0.08
M=0
M=0.21
Probability Density
.6
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
y
help("plot_density")

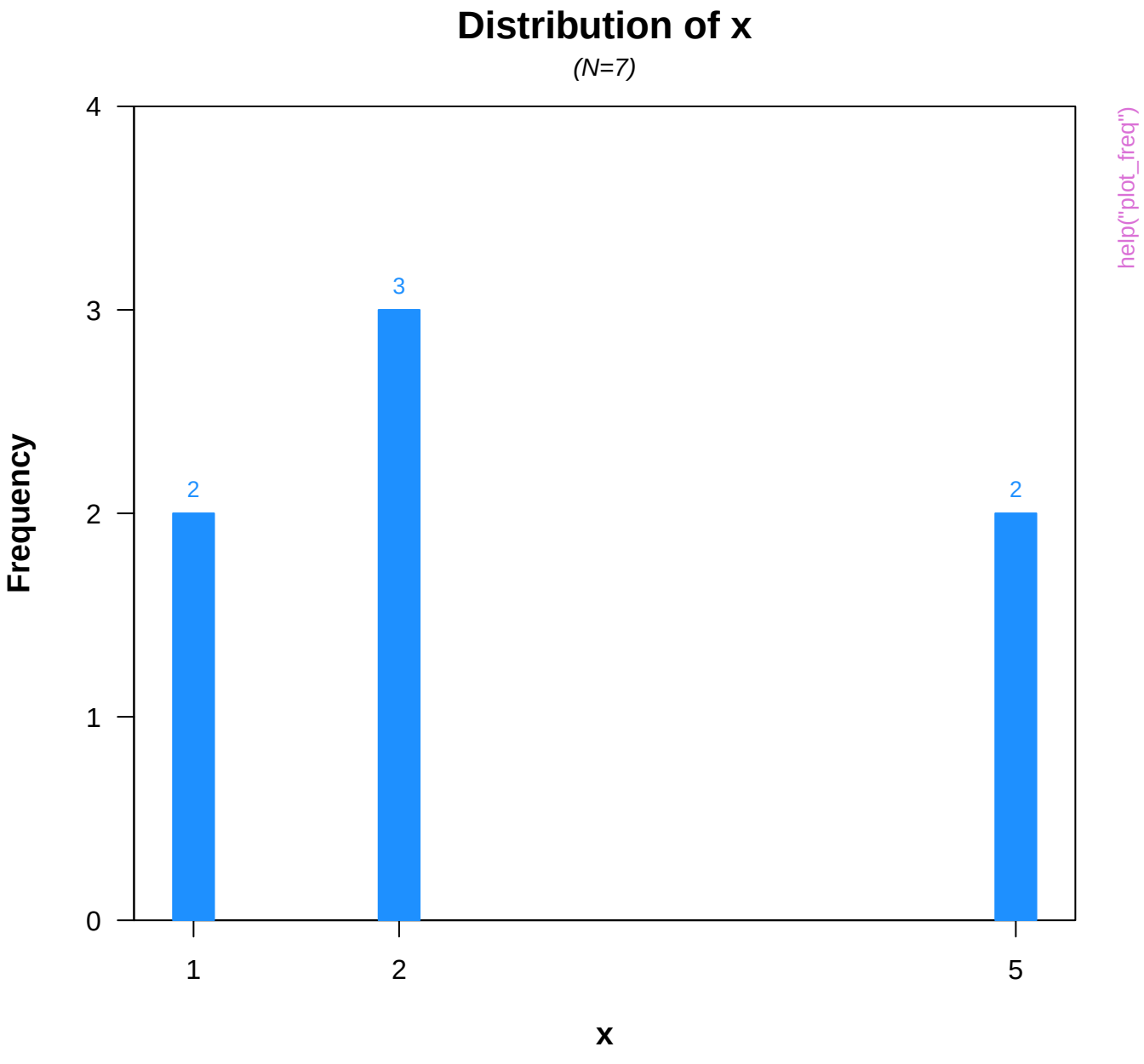

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.05
M=−0.02
Probability Density
.6
.5
.4
.3
.2
.1
0
−3
−2
−1
0
1
2
3
value
help("plot_density")

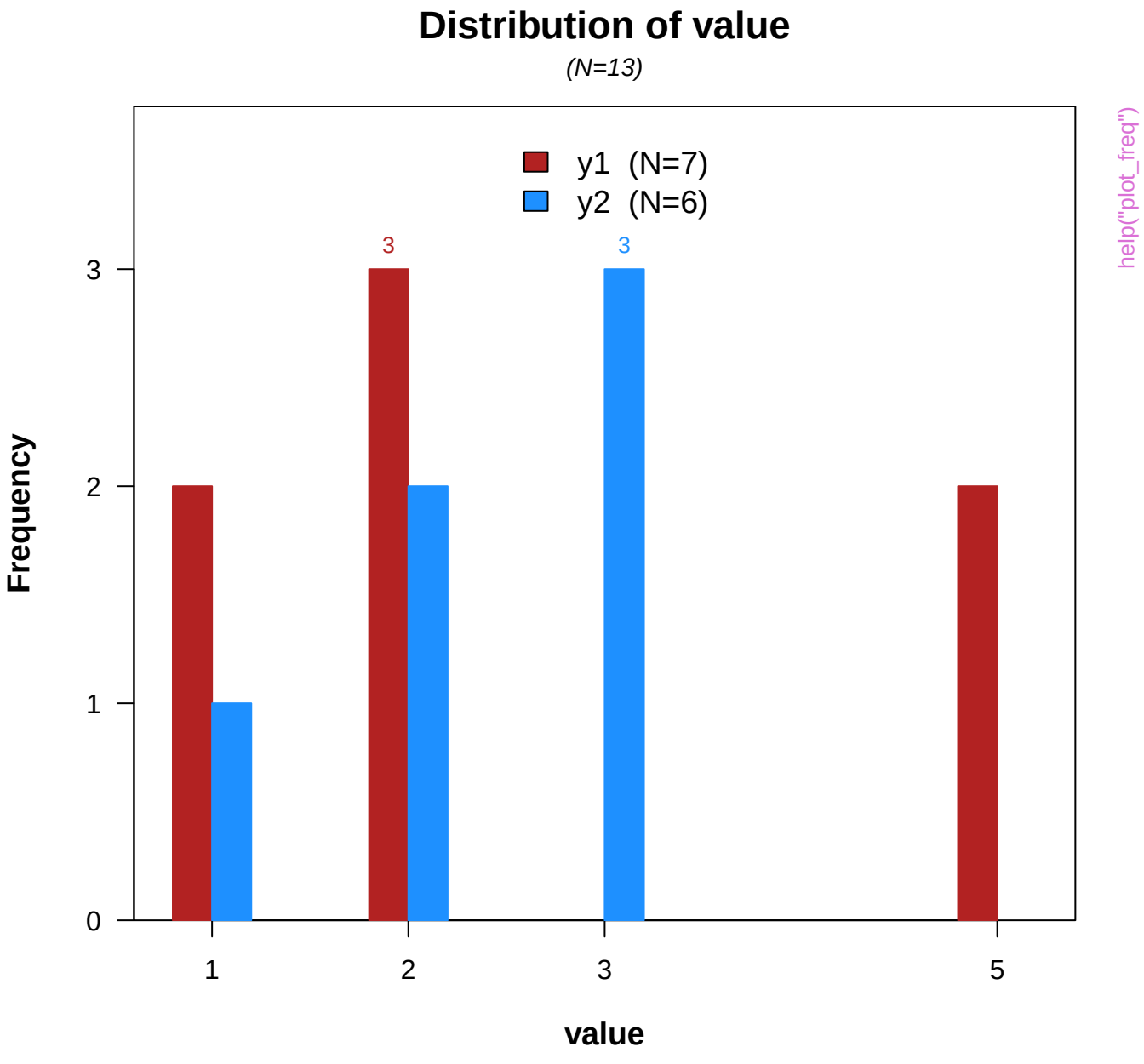

Comparing Distribution of 'value' by 'group'
(n=100)
A
(n=50)
B
(n=50)
M=−0.05
M=−0.02
Probability Density
.6
.5
.4
.3
.2
.1
0
−3
−2
−1
0
1
2
3
value
help("plot_density")

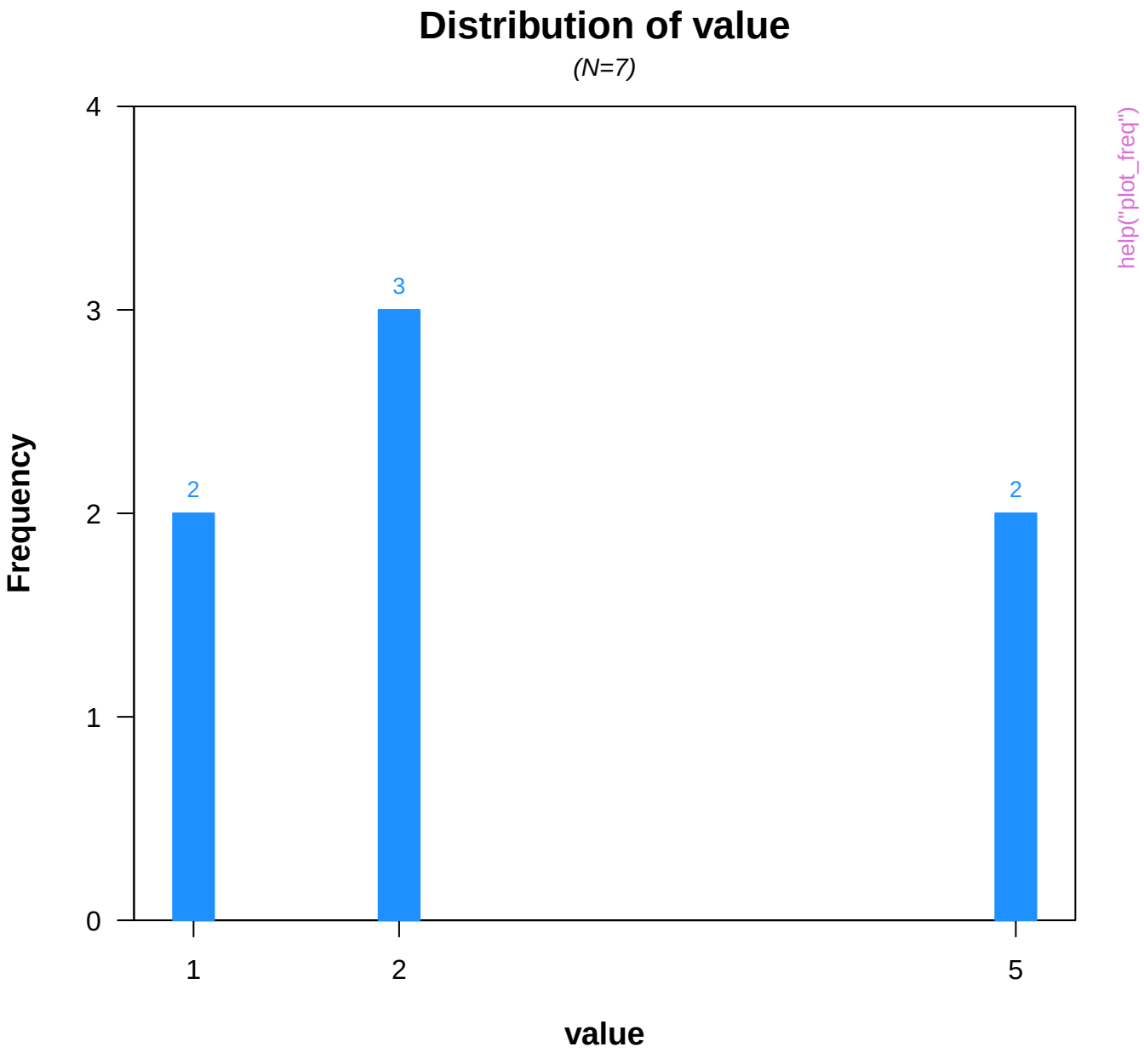

Comparing Distribution of 'value' by 'group'
(n=100)
y1
(n=50)
y2
(n=50)
M=−0.03
M=1.09
Probability Density
.5
.4
.3
.2
.1
0
−4
−2
0
2
4
value
help("plot_density")

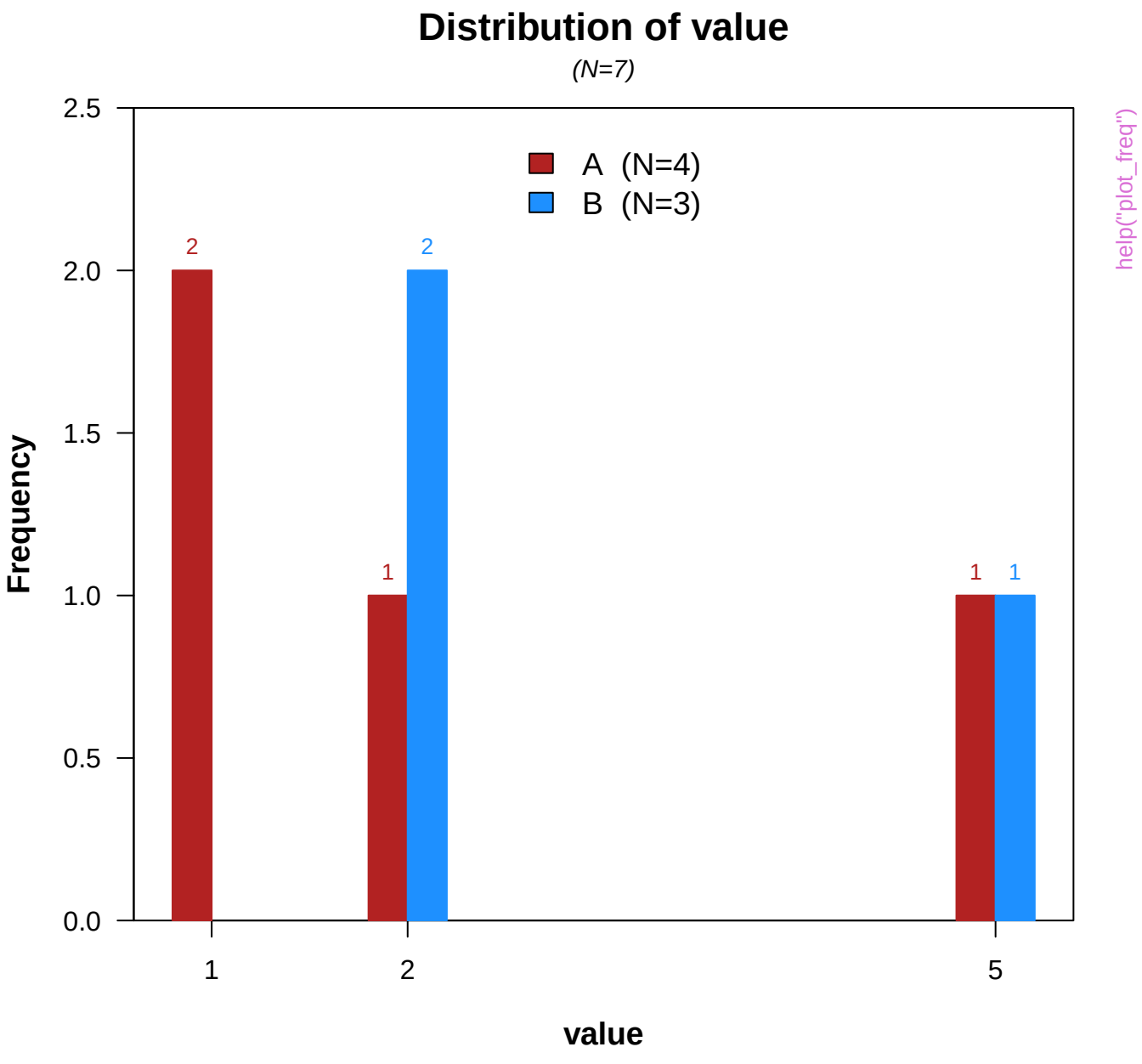

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x
help("plot_freq")

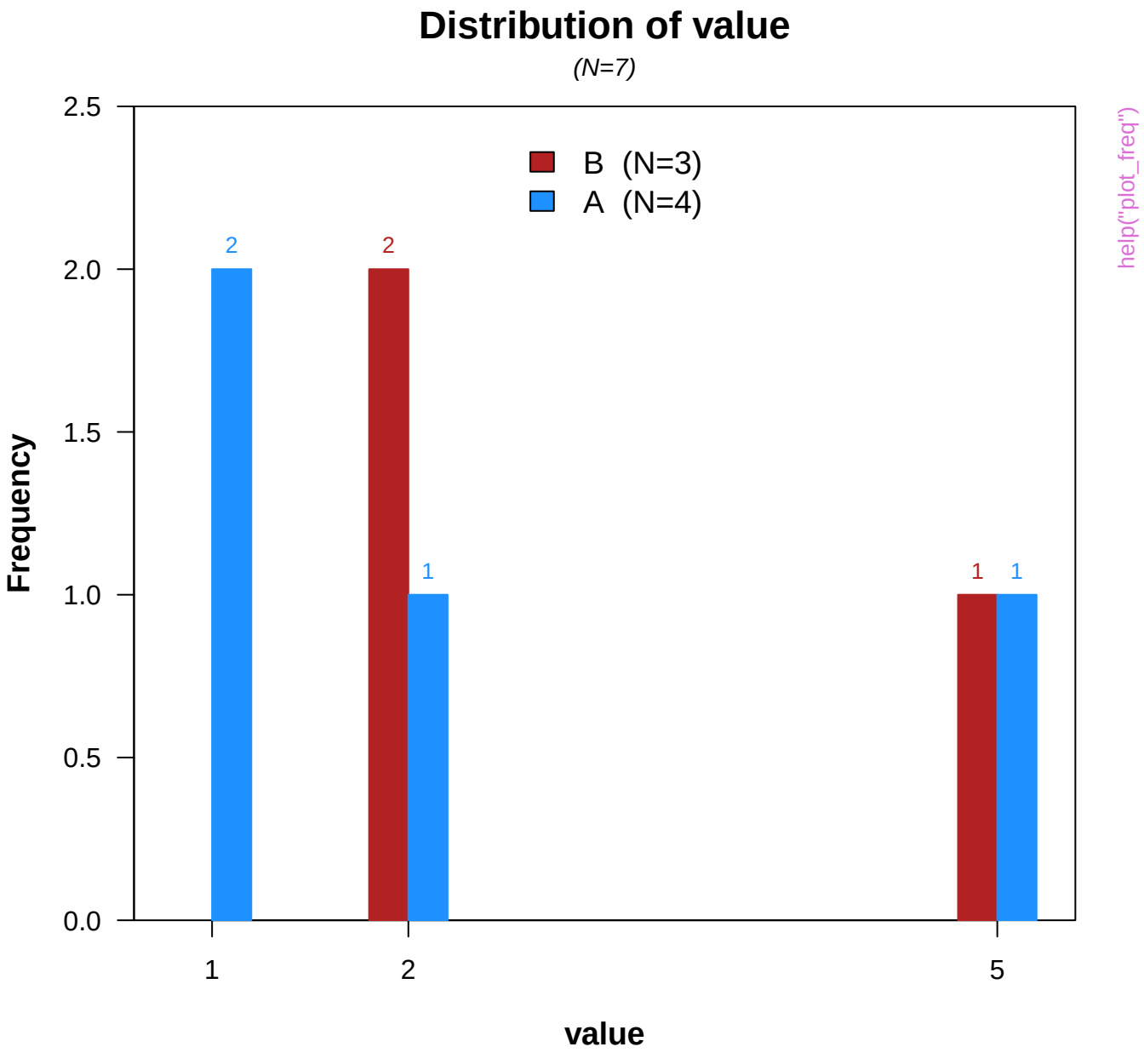

Distribution of x
(N=7)
Frequency
Value
0
1
2
3
4
5
6
7
1
2
5
2
3
2
help("plot_freq")

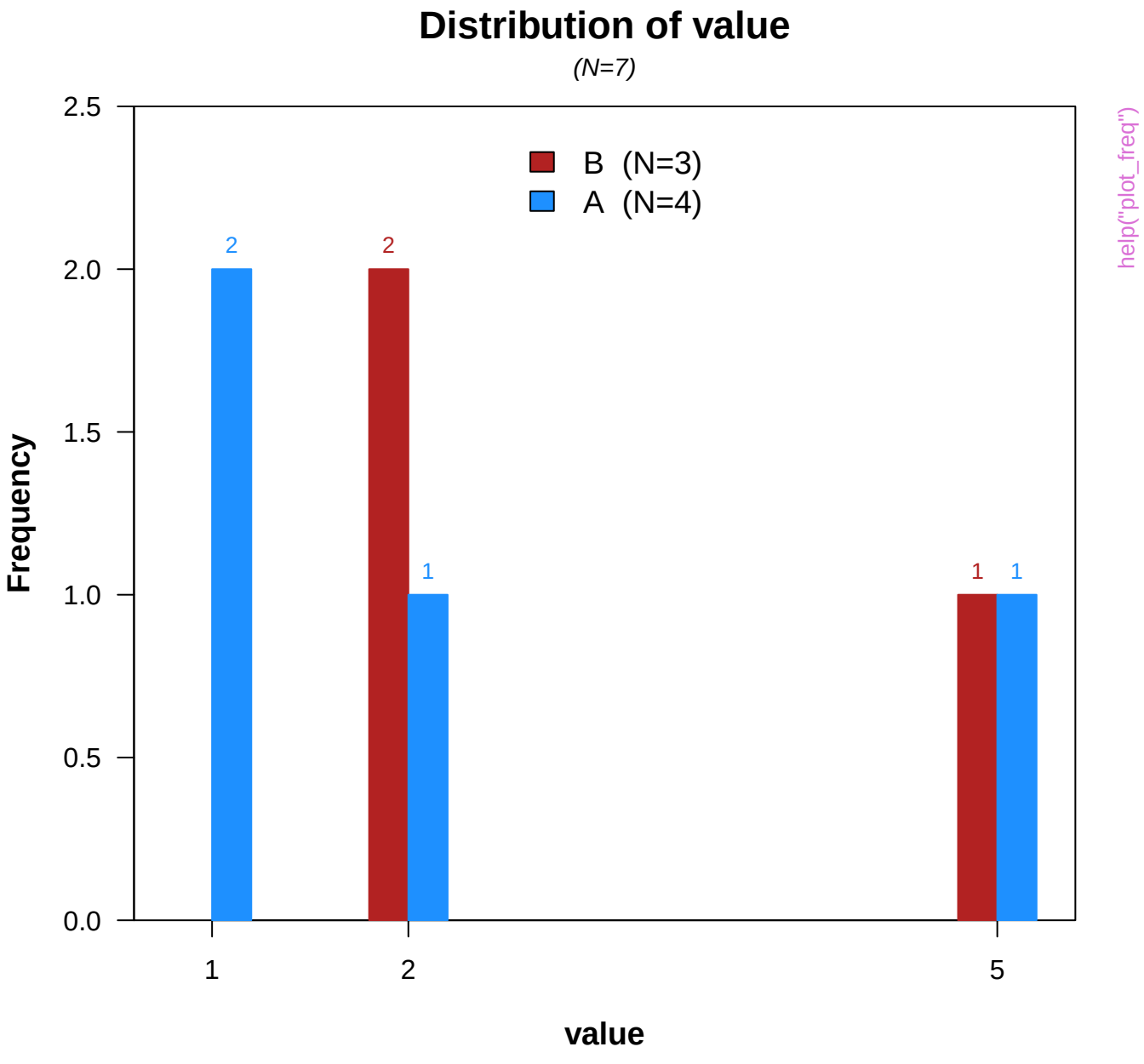

Distribution of x
(N=7)
Frequency
4
3
2
1
0
2
3
2
1
2
5
x
help("plot_freq")

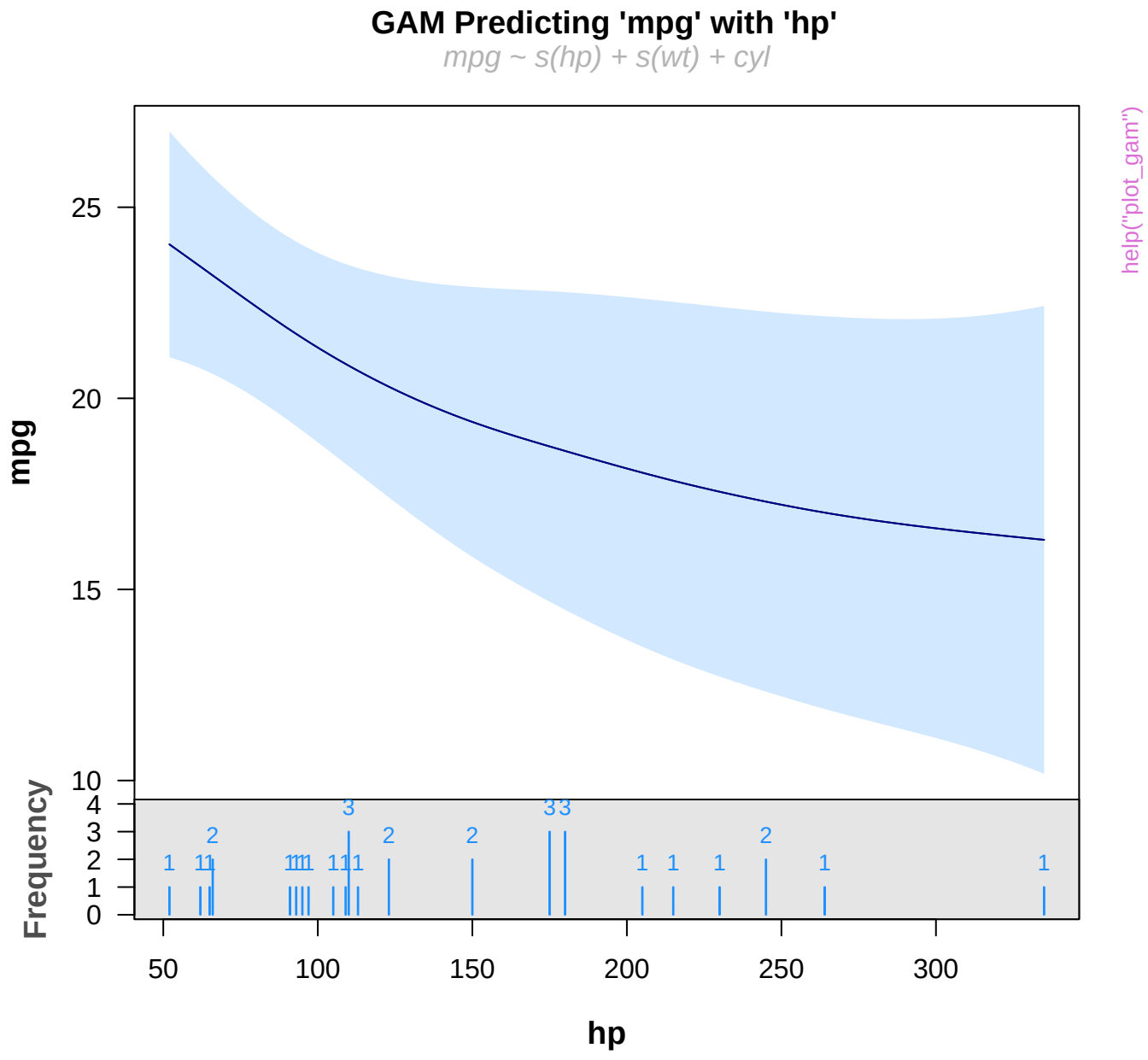

Distribution of value
(N=13)
y1 (N=7)
y2 (N=6)
3
3
Frequency
0
1
2
3
1
2
3
5
value
help("plot_freq")

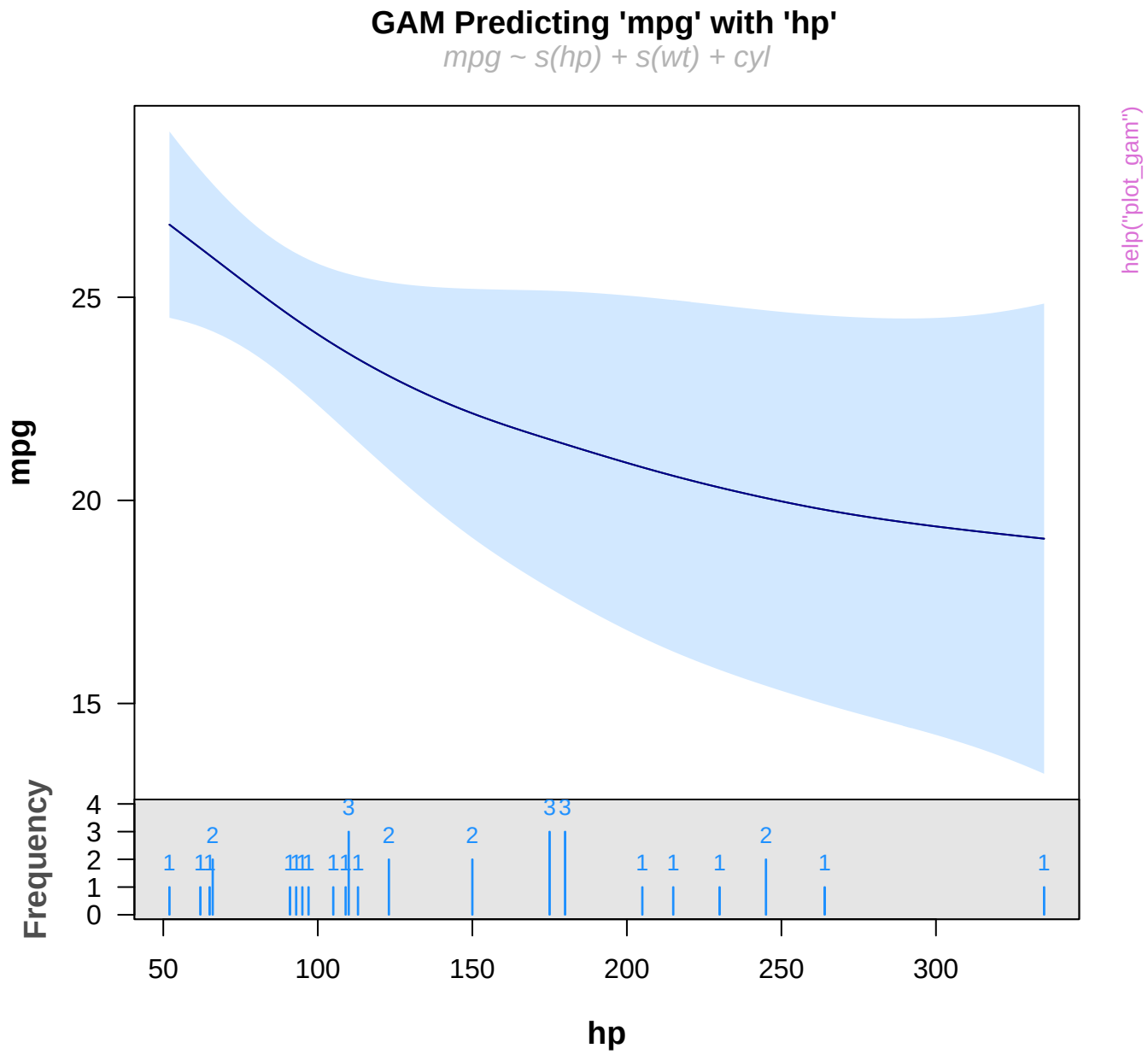

Distribution of value
(N=7)
Frequency
value
0
1
2
3
4
1
2
5
2
3
2
help("plot_freq")

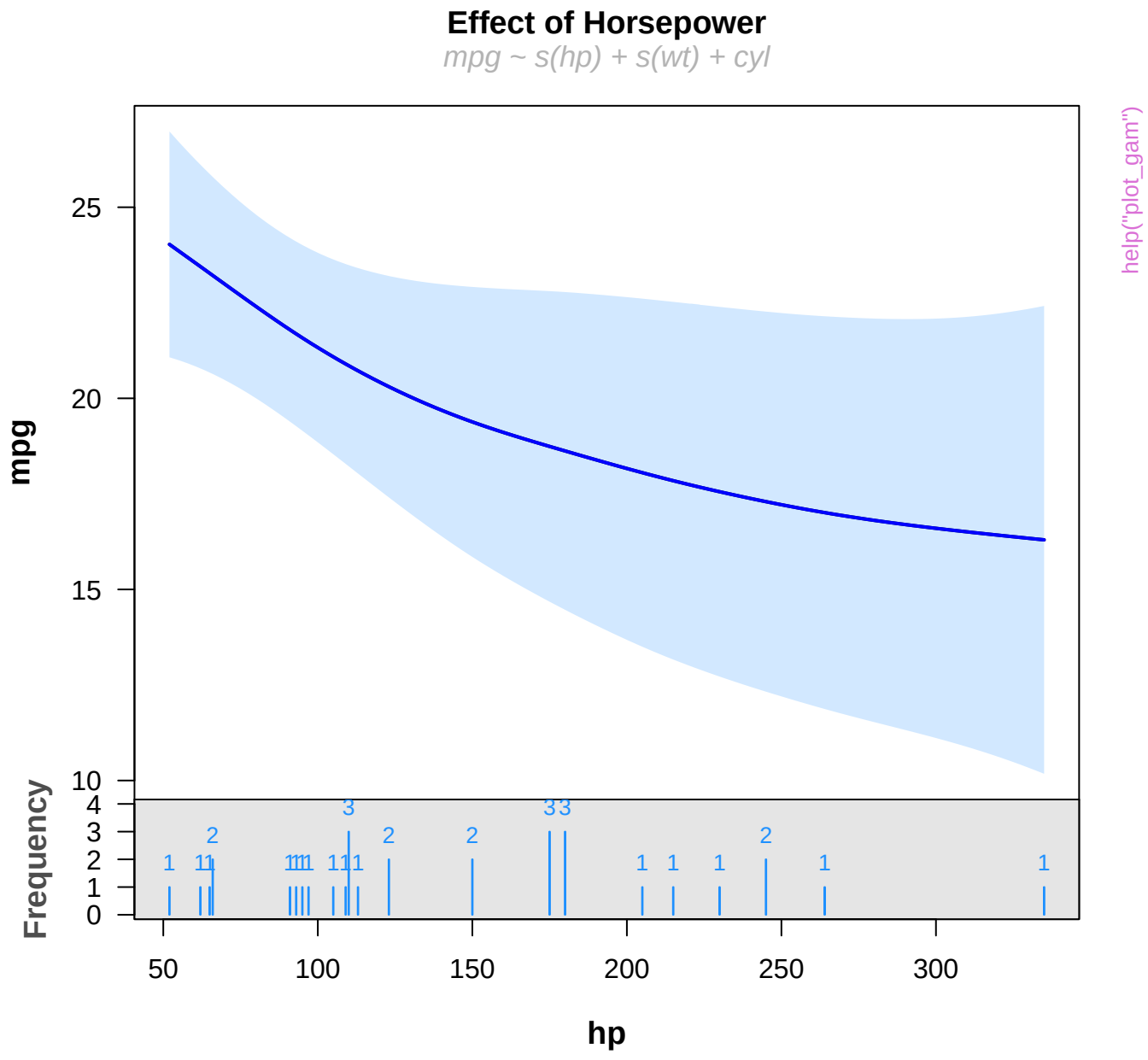

Distribution of value
(N=7)
A (N=4)
B (N=3)
Frequency
value
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
2
2
1
1
1
help("plot_freq")

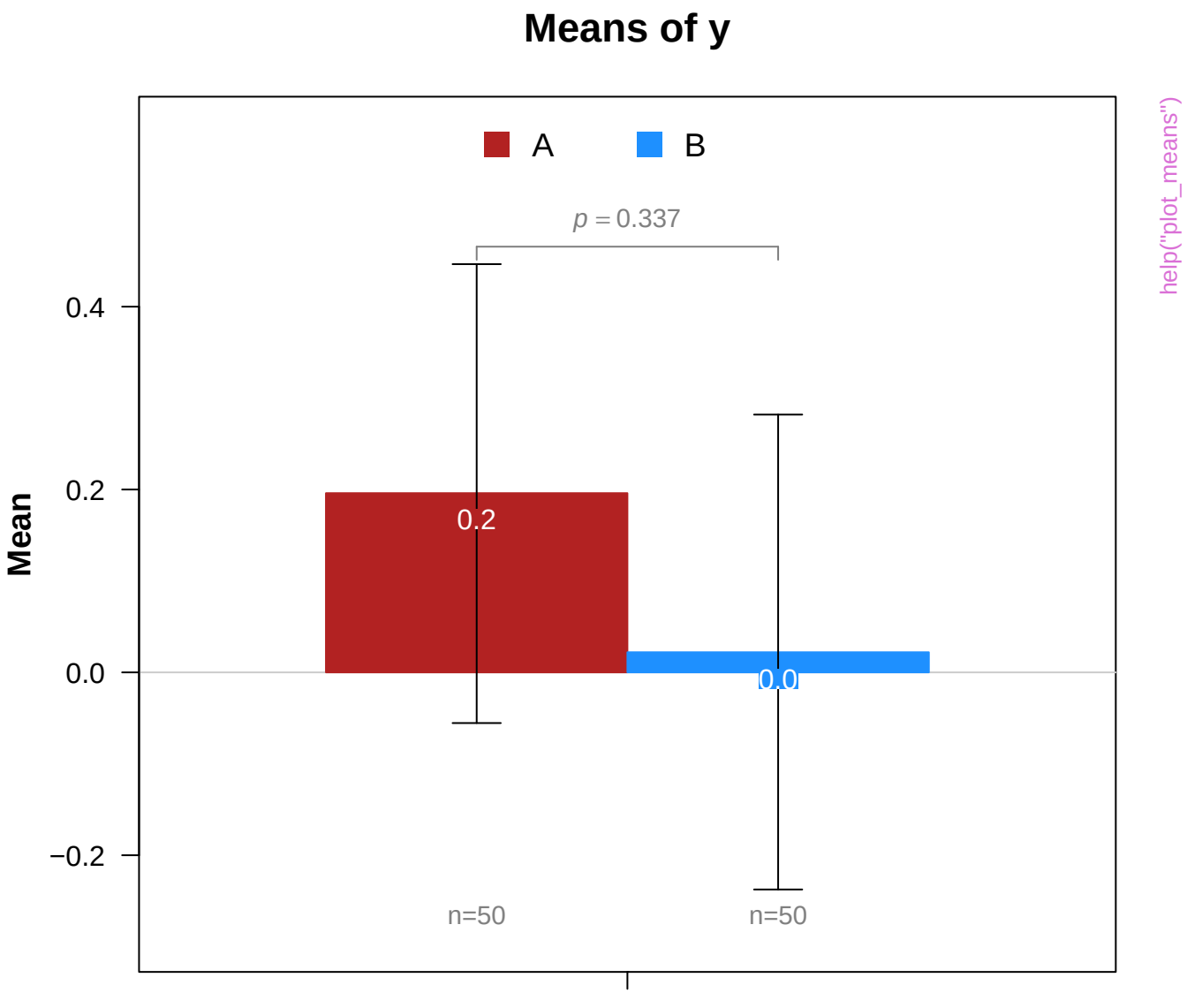

Distribution of value
(N=7)
B (N=3)
A (N=4)
Frequency
value
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
2
2
1
1
1
help("plot_freq")

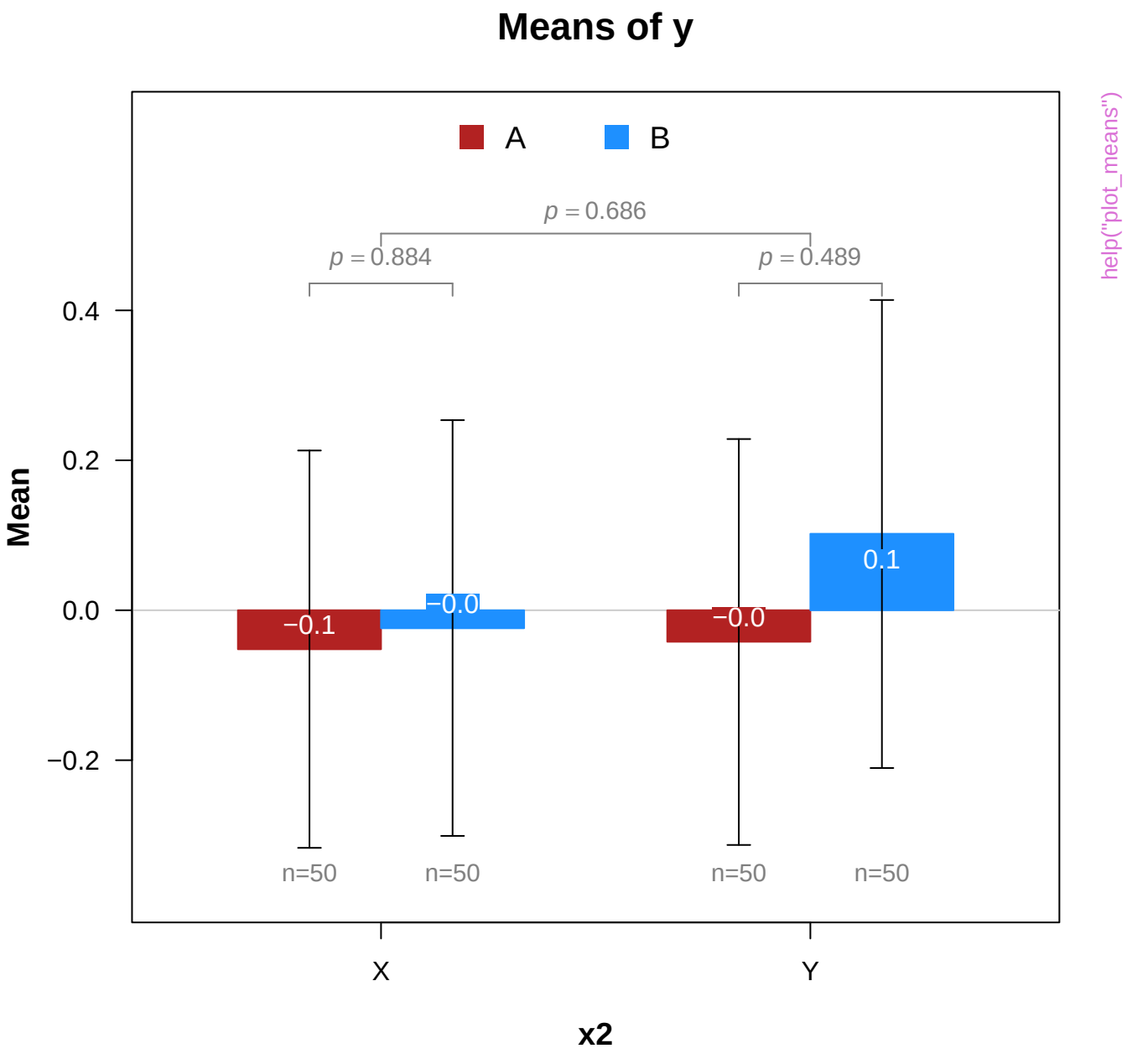

Distribution of value
(N=7)
B (N=3)
A (N=4)
Frequency
value
2.5
2.0
1.5
1.0
0.5
0.0
1
2
5
2
2
1
1
1
help("plot_freq")

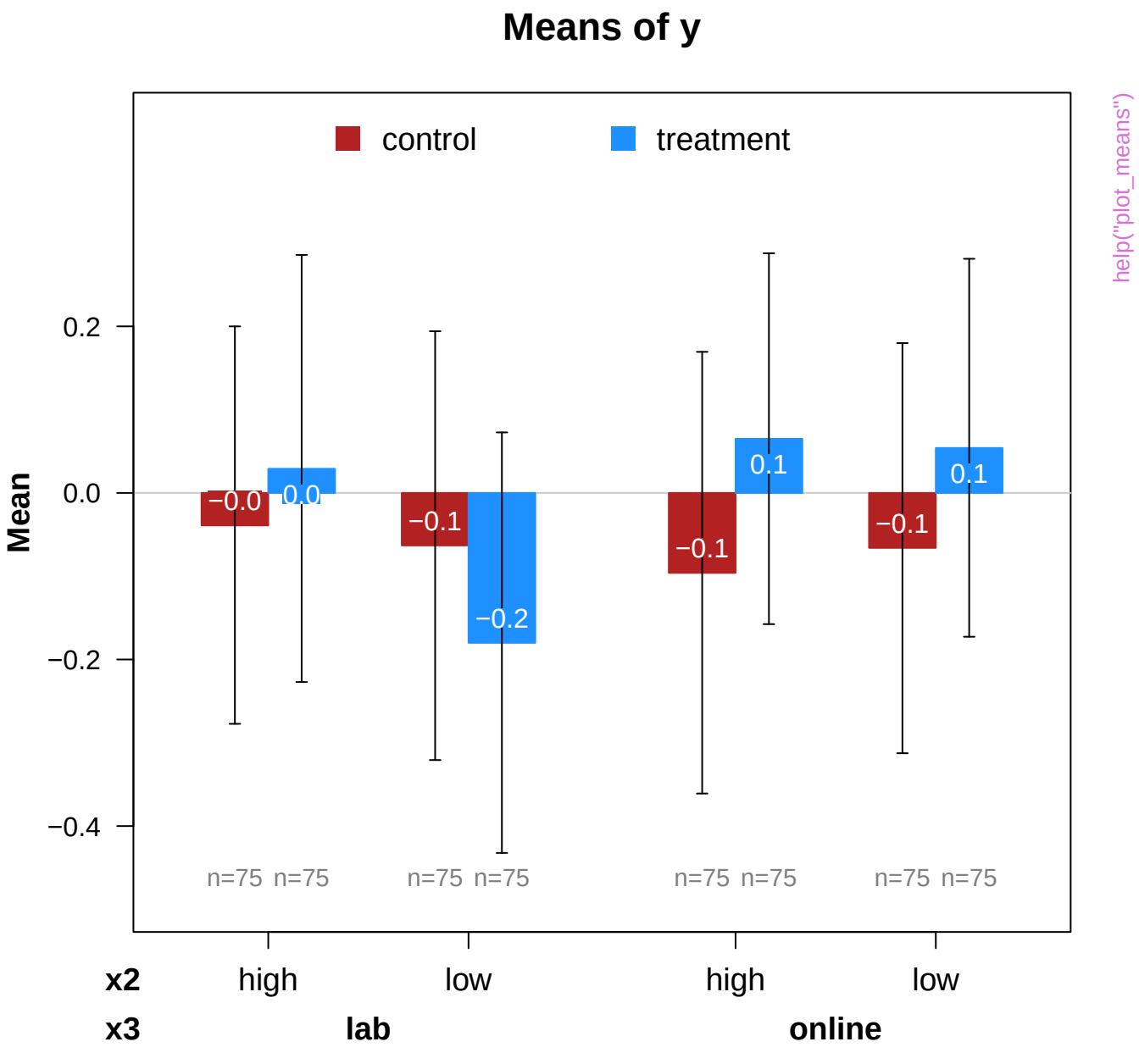

GAM Predicting 'mpg' with 'hp'
mpg ~ s(hp) + s(wt) + cyl
mpg
25
20
15
10
Frequency
4
3
2
1
0
50
100
150
200
250
300
hp
help("plot_gam")

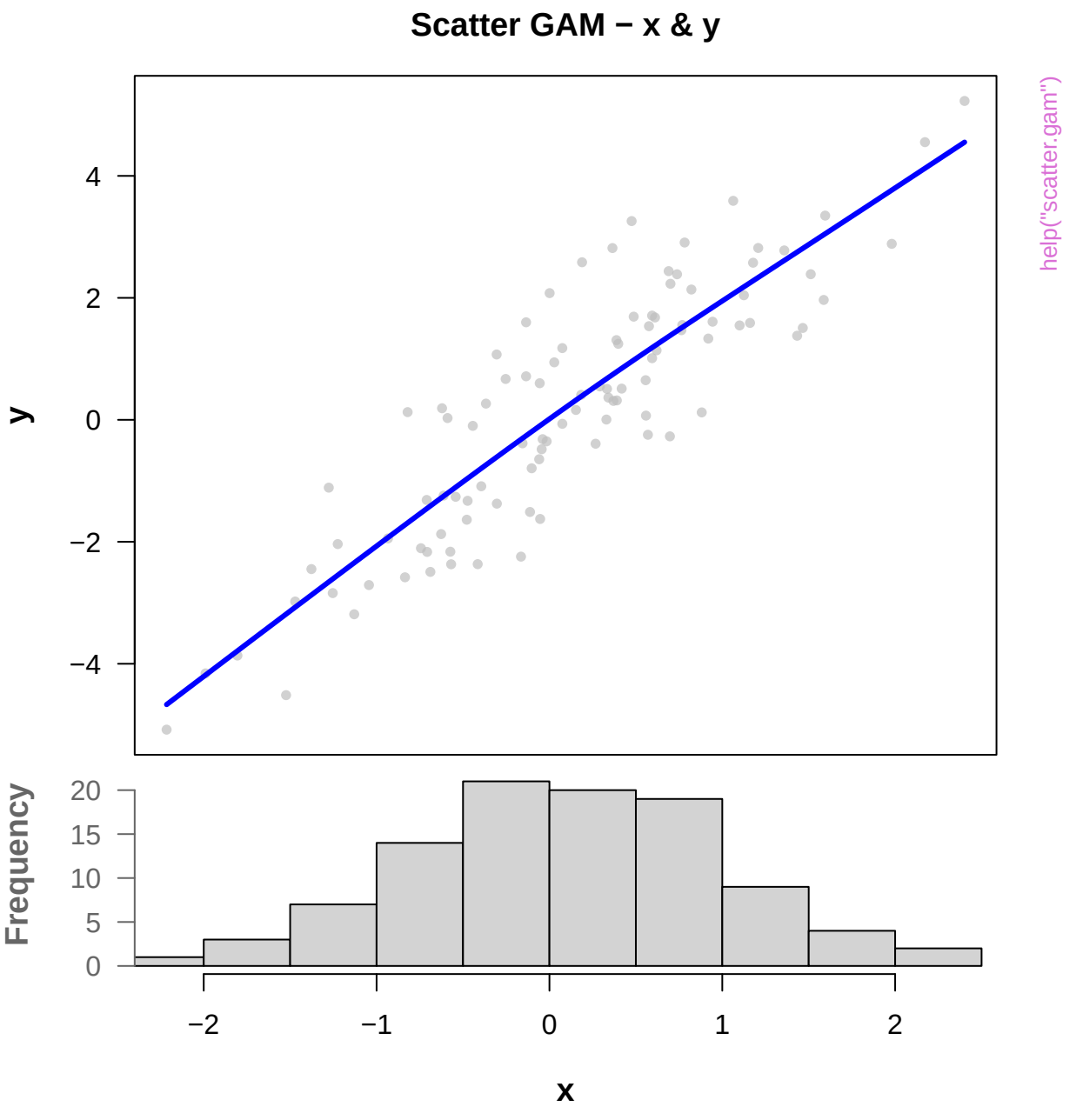

GAM Predicting 'mpg' with 'hp'
mpg ~ s(hp) + s(wt) + cyl
mpg
25
20
15
Frequency
4
3
2
1
0
50
100
150
200
250
300
hp
help("plot_gam")

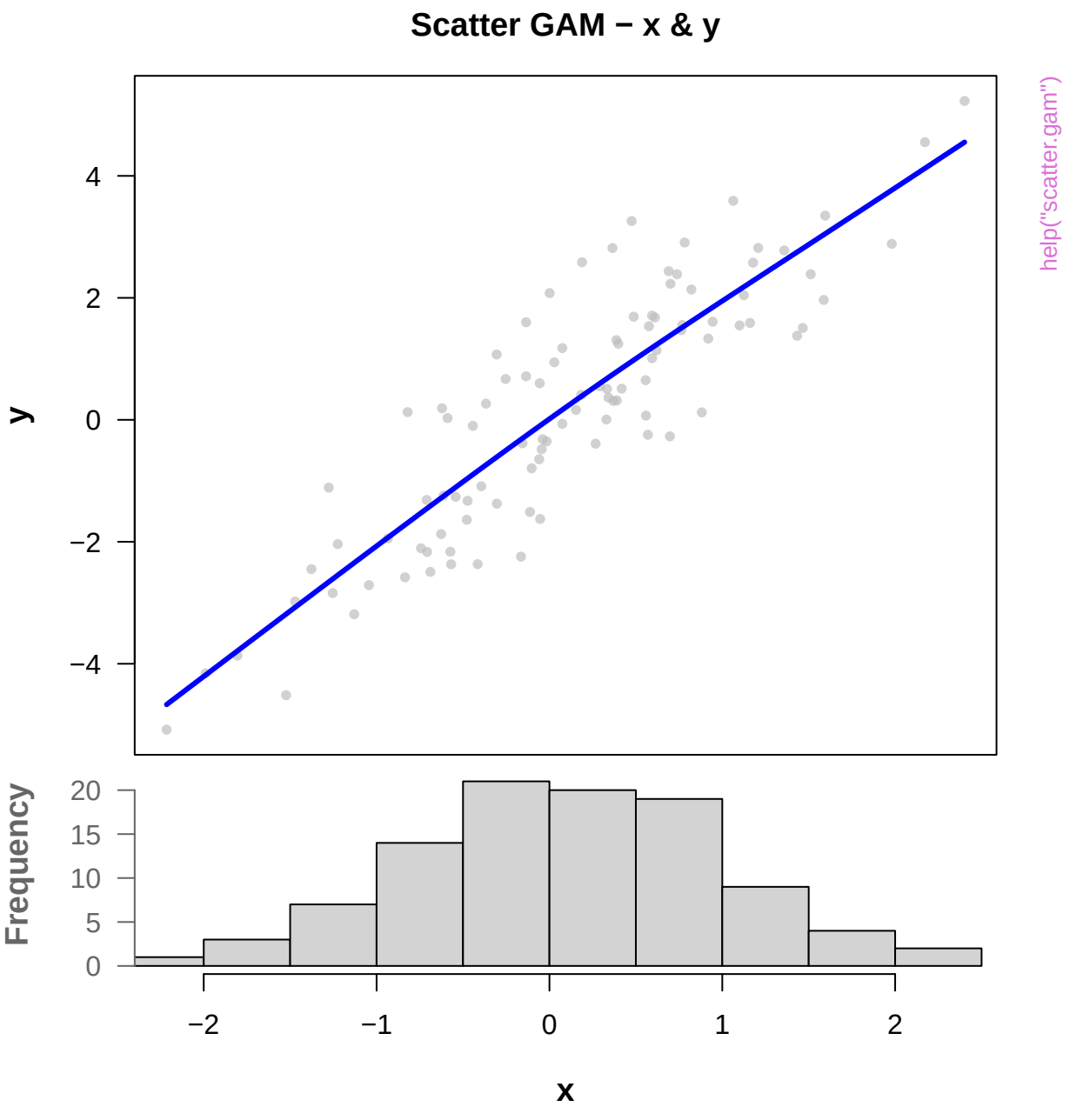

Effect of Horsepower
mpg ~ s(hp) + s(wt) + cyl
mpg
25
20
15
10
Frequency
4
3
2
1
0
50
100
150
200
250
300
hp
help("plot_gam")

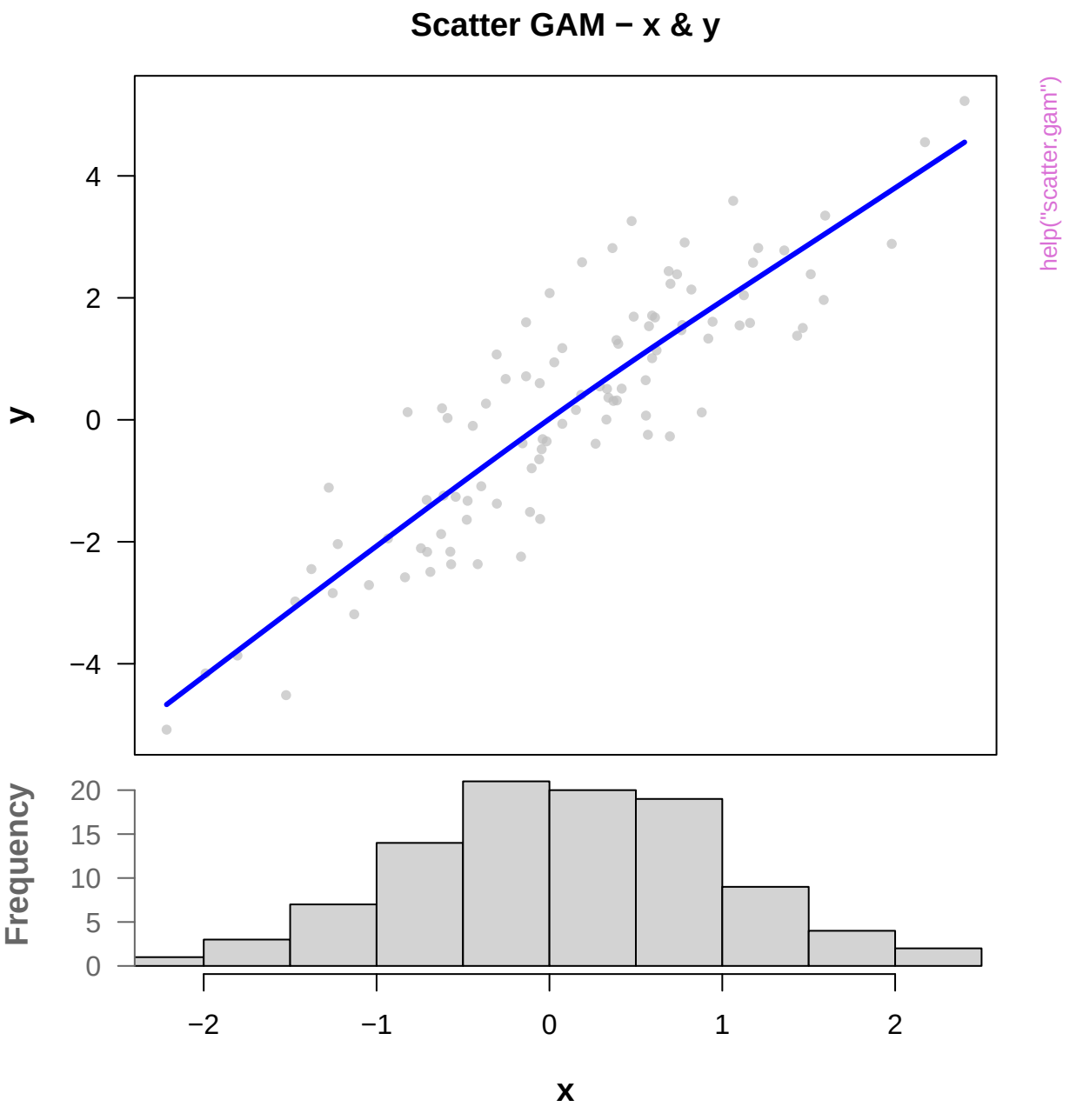

Means of y
A
B
p = 0.337
0.4
0.2
0.0
−0.2
Mean
0.2
0.0
n=50
n=50
help("plot_means")

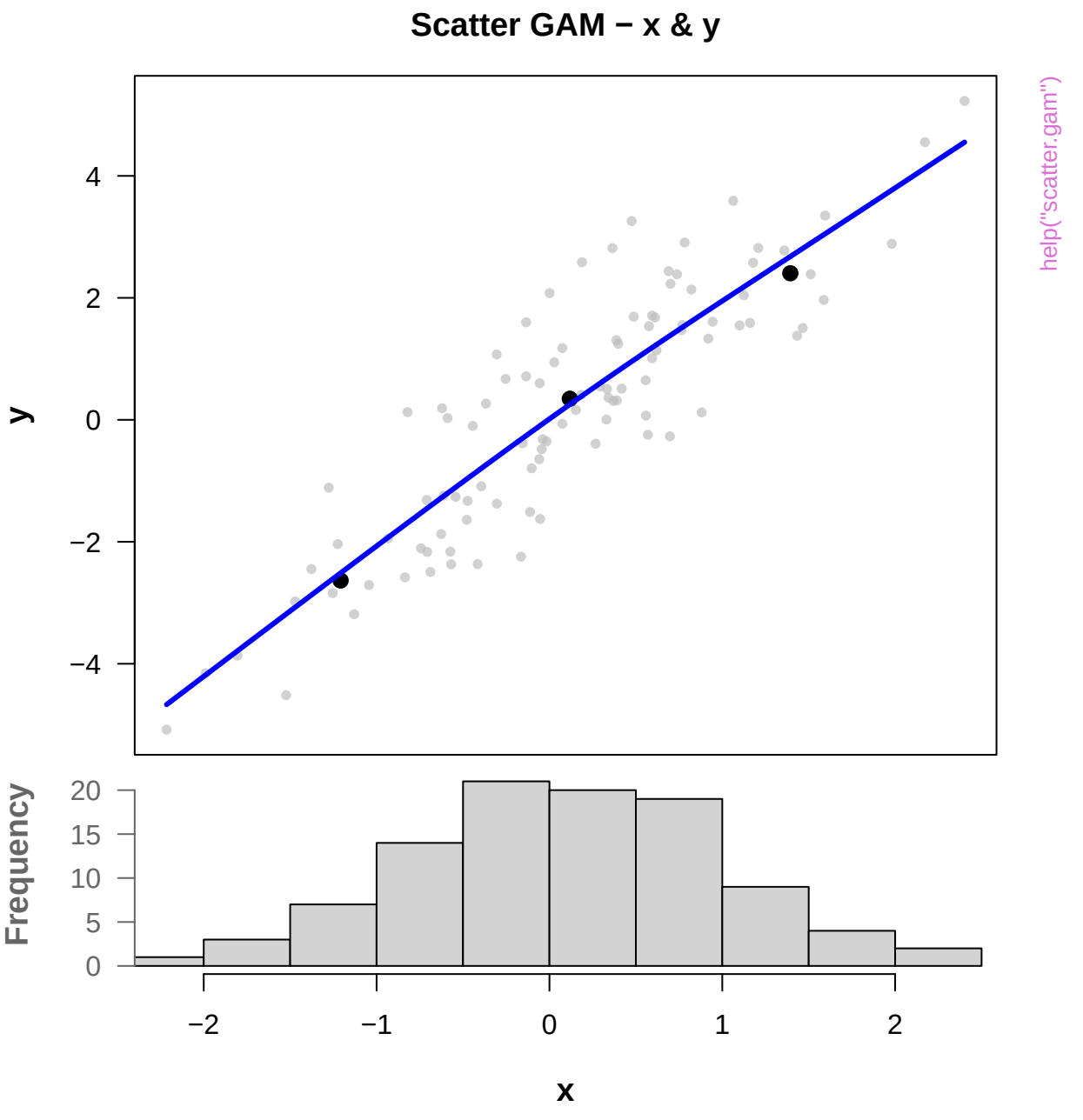

Means of y
A
B
p = 0.686
p = 0.884
p = 0.489
0.4
0.2
0.0
−0.2
Mean
−0.1
−0.0
−0.0
0.1
n=50
n=50
n=50
n=50
X
Y
x2
help("plot_means")

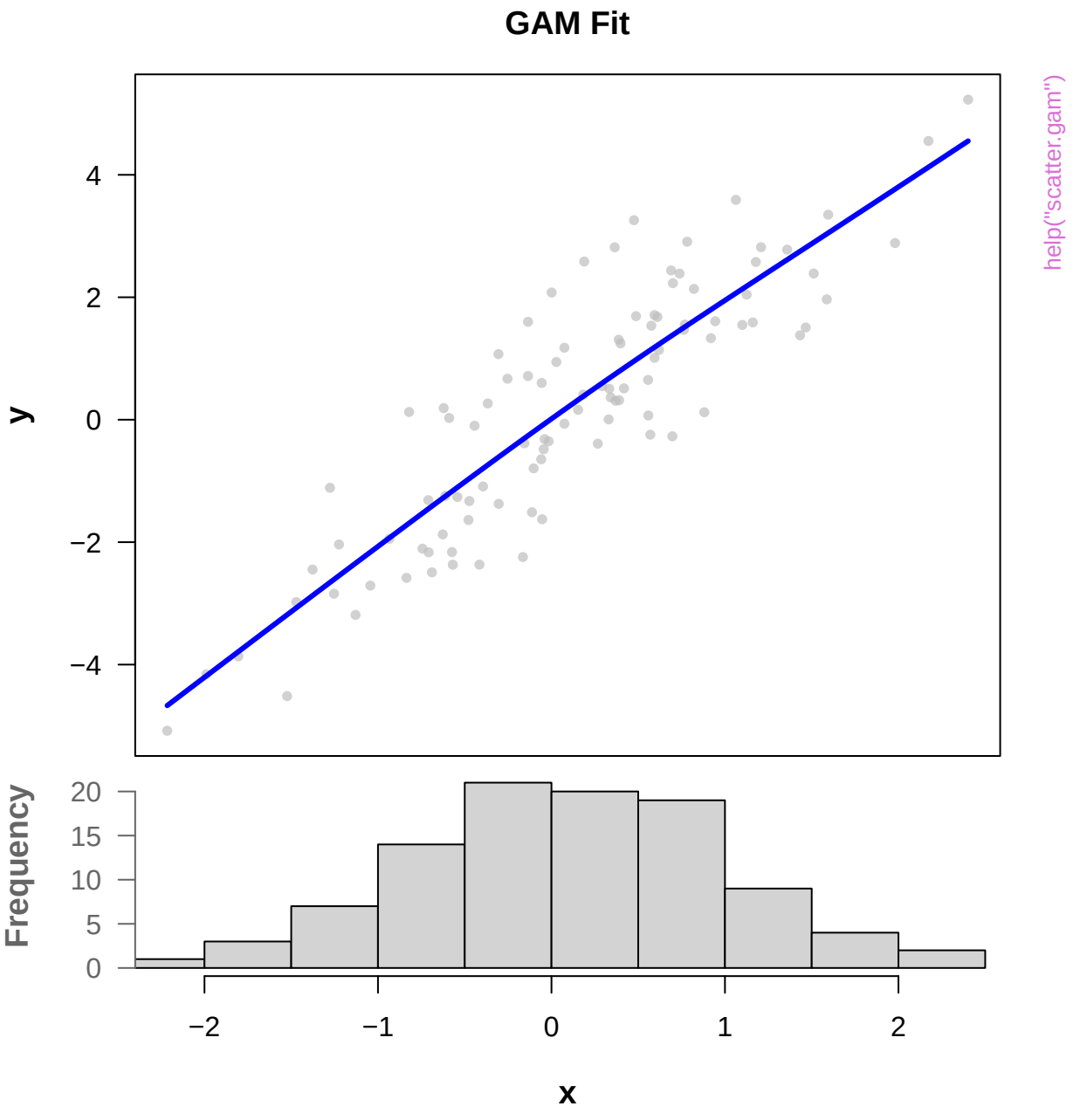

Means of y
control
treatment
Mean
0.2
0.0
−0.2
−0.4
−0.0
0.0
−0.1
−0.2
−0.1
0.1
−0.1
0.1
n=75 n=75
n=75 n=75
n=75 n=75
n=75 n=75
x2
high
low
high
low
x3
lab
online
help("plot_means")

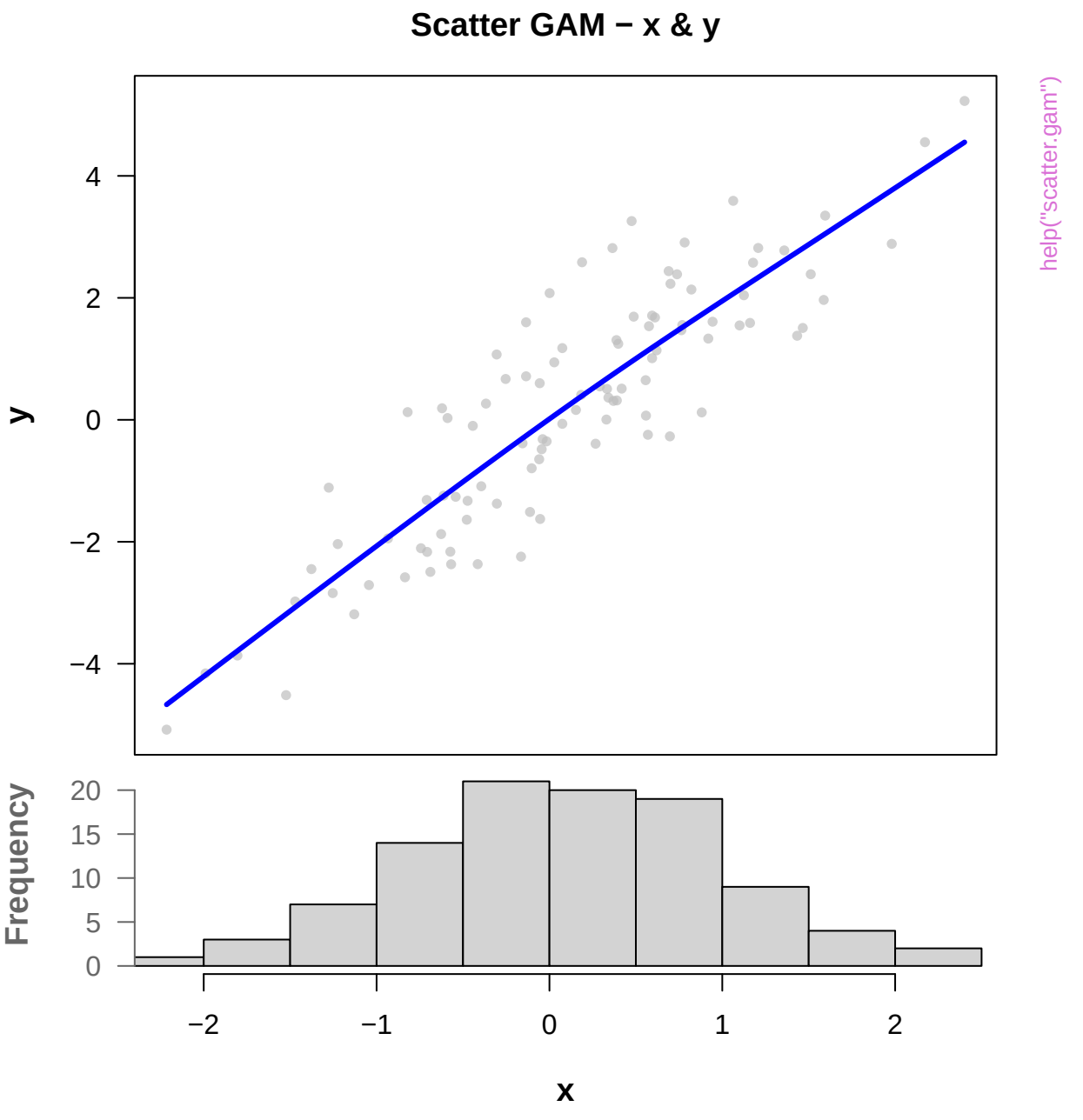

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")

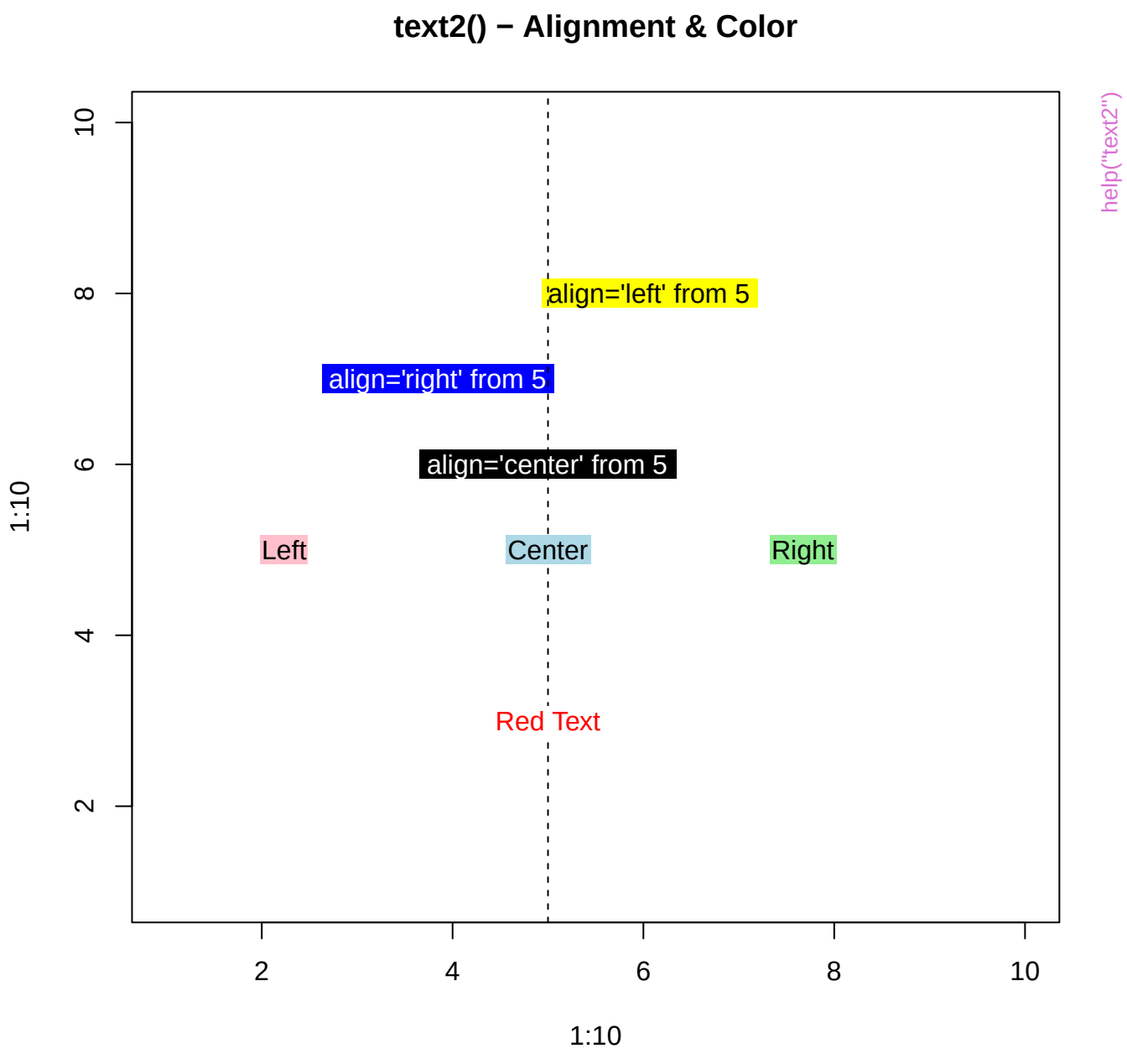

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")

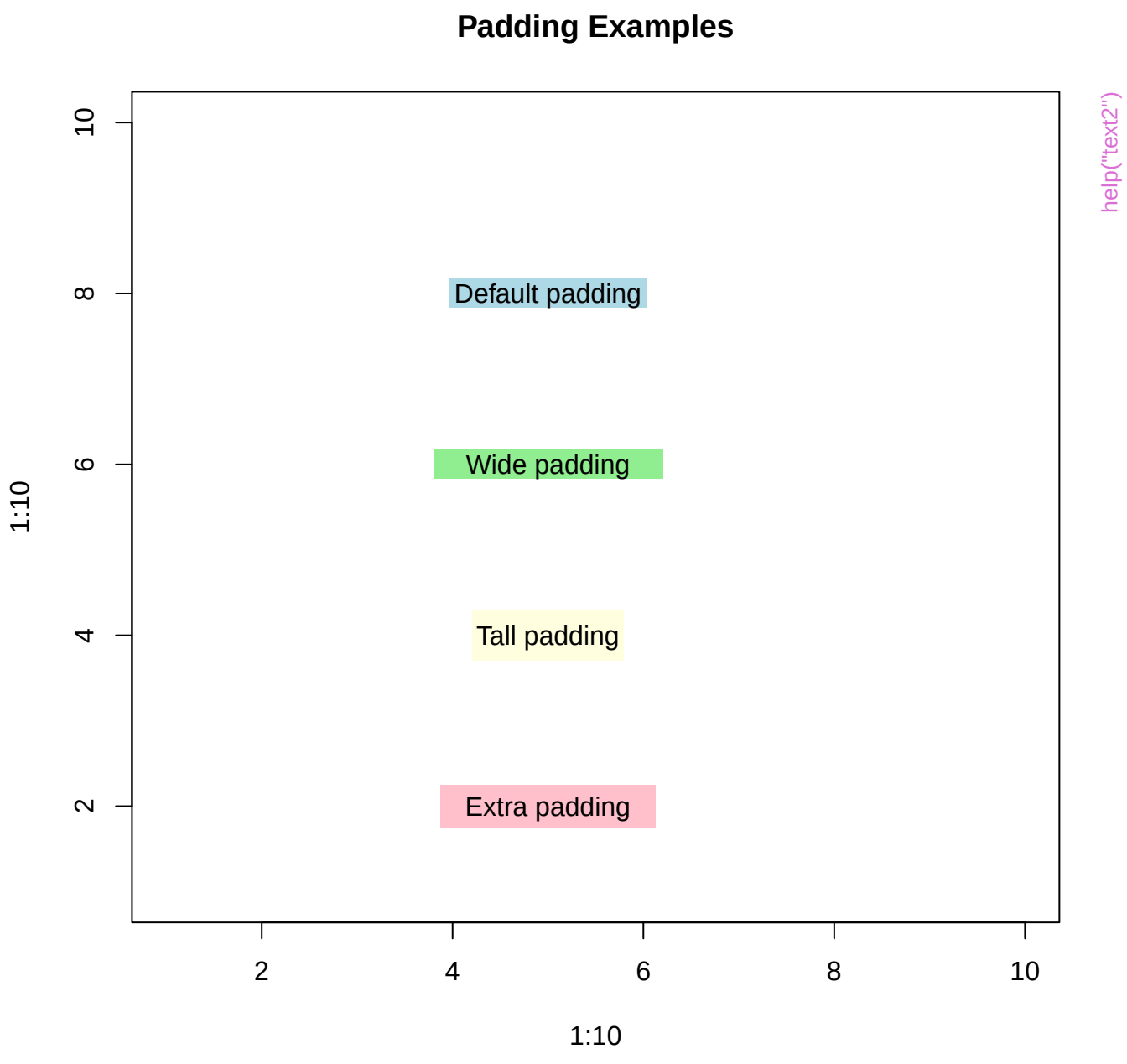

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")

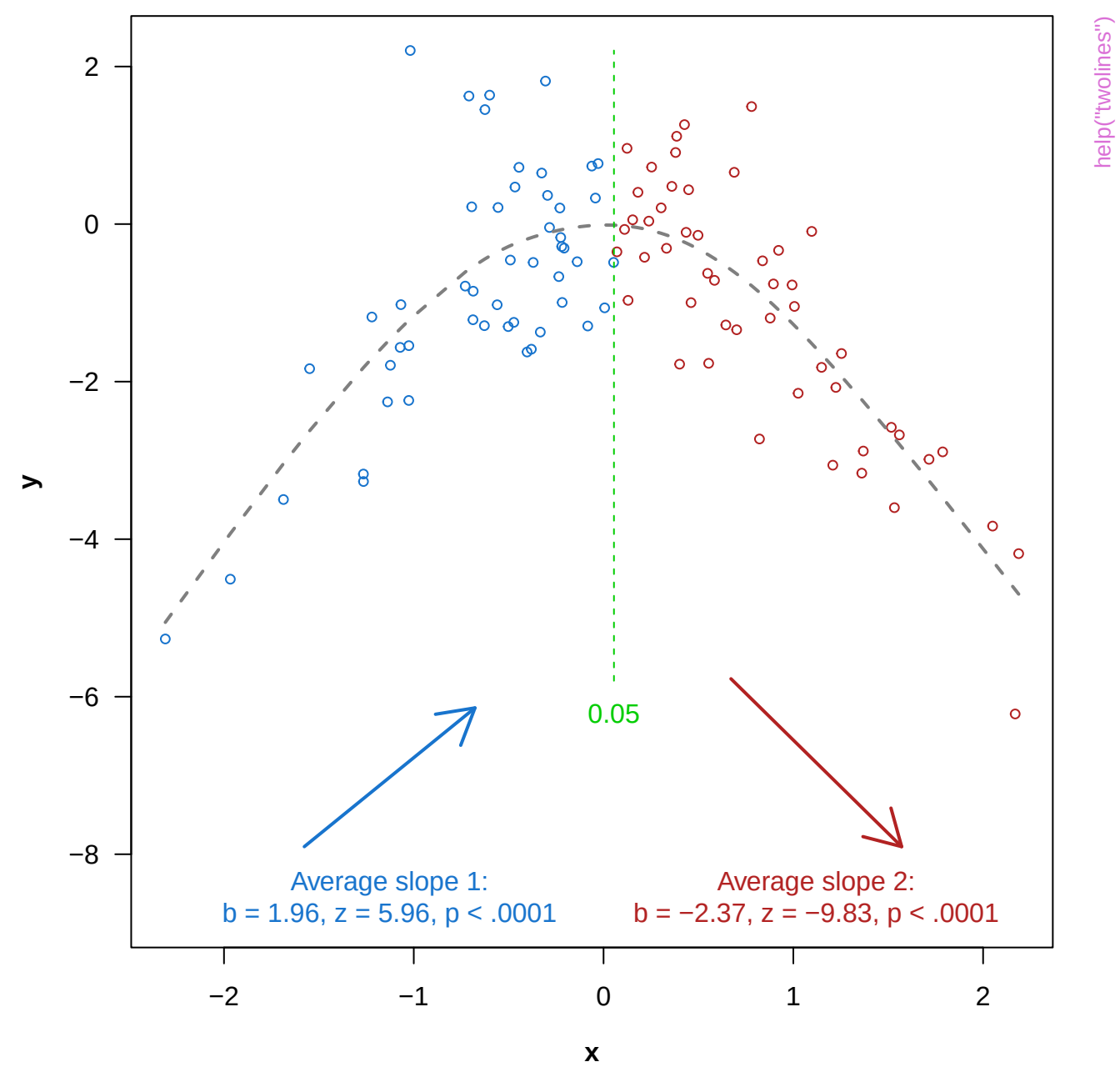

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")

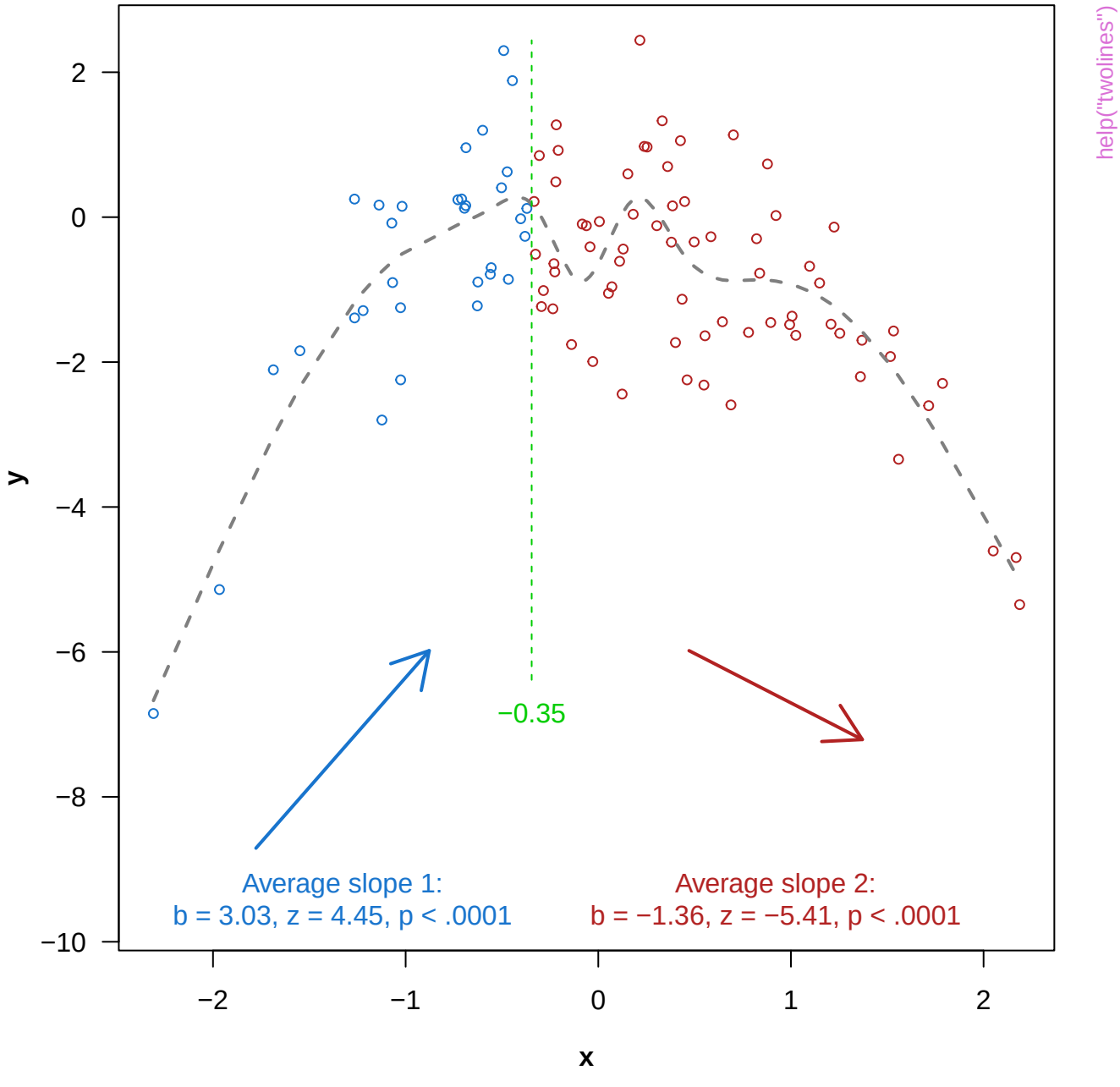

GAM Fit
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")

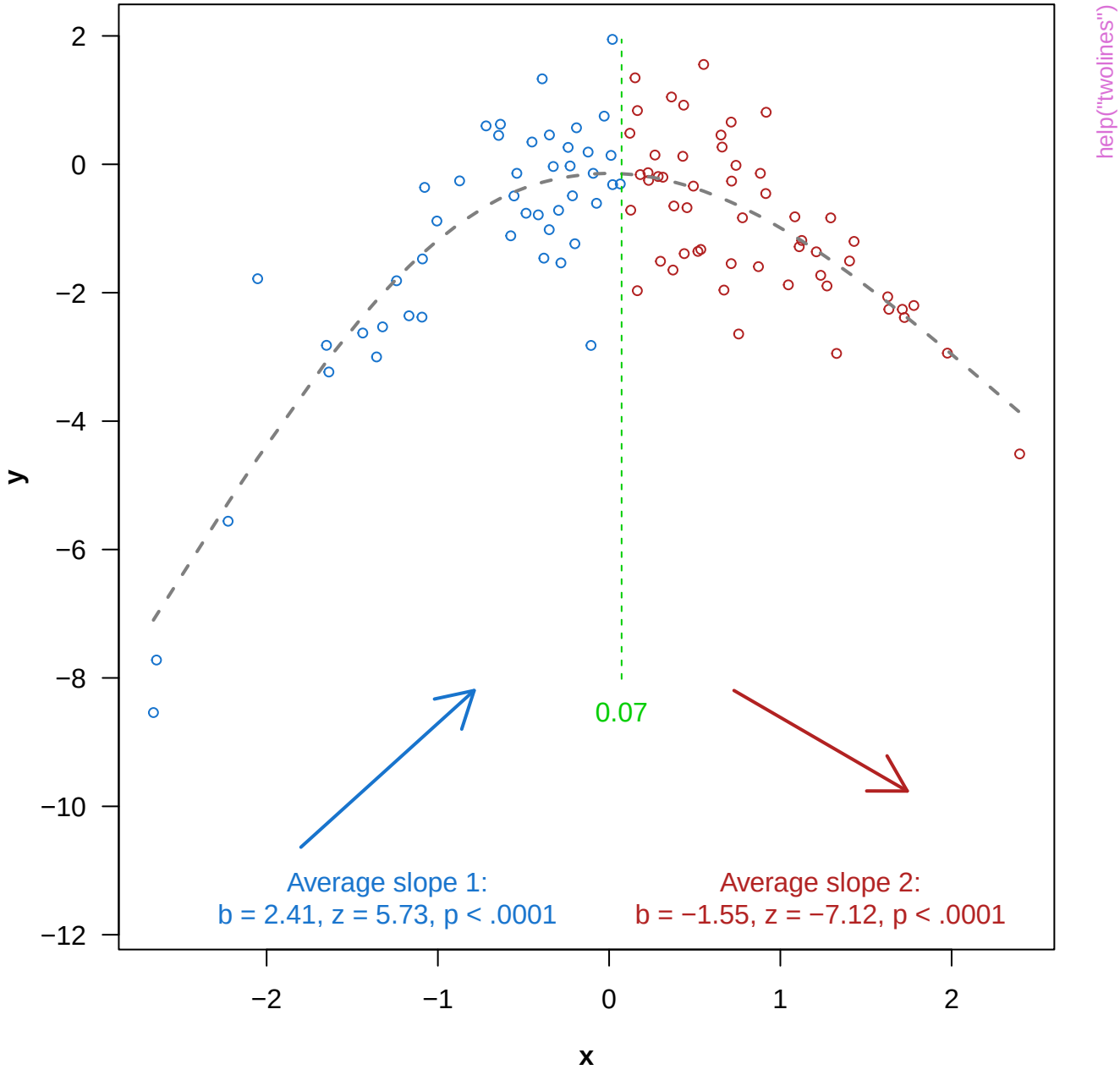

Scatter GAM − x & y
y
4
2
0
−2
−4
Frequency
20
15
10
5
0
−2
−1
0
1
2
x
help("scatter.gam")


text2() − Alignment & Color
help("text2")
align='left' from 5
align='right' from 5
align='center' from 5
Left
Center
Right
Red Text
1:10
1:10
2
4
6
8
10


Padding Examples
Default padding
Wide padding
Tall padding
Extra padding
1:10
1:10
2
4
6
8
10
2
4
6
8
10
help("text2")


help("twolines")
2
0
−2
−4
−6
−8
y
−2
−1
0
1
2
x
0.05
Average slope 1:
b = 1.96, z = 5.96, p < .0001
Average slope 2:
b = −2.37, z = −9.83, p < .0001


help("twolines")
y
x
2
0
−2
−4
−6
−8
−10
−2
−1
0
1
2
−0.35
Average slope 1:
b = 3.03, z = 4.45, p < .0001
Average slope 2:
b = −1.36, z = −5.41, p < .0001


help("twolines")
2
0
−2
−4
−6
−8
−10
−12
y
−2
−1
0
1
2
x
0.07
Average slope 1:
b = 2.41, z = 5.73, p < .0001
Average slope 2:
b = −1.55, z = −7.12, p < .0001